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Zoo103 Final Term Handouts 2022

LECTURES 109 to 218

M Zaman

(0304-4756496)

MCQS & SUBJECTIVE HIGHLIGHTED

(Lec 109 to 130 Most imporatant_(MCQs and SHORT QESTION 2, 3 Marks) & next remember the ALL morphologies, Classifications and Lifecycles_(MOST IMPORTANT FOR 5 marks Questions) lec_194 to next Most Important as a MCQs and Short Questions)

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Module# 109 Sampling Methods

(All sampling methods are very important)

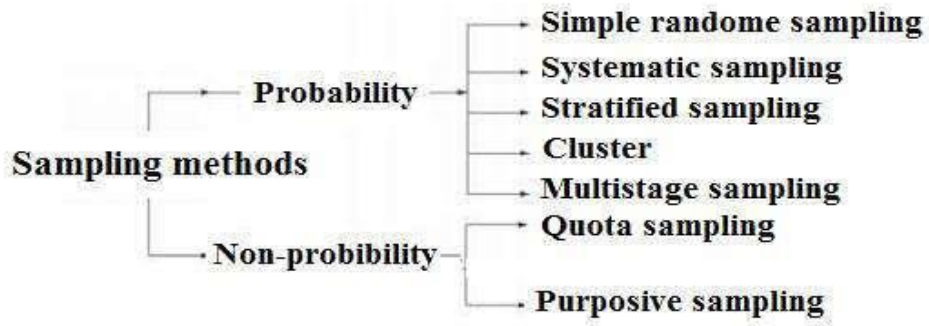
(Remember their definitions and Examples)

- Probability Sampling
- Non-Probability Sampling Probability Sampling
- Random Sampling
- Systematic Sampling

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- Stratified sampling
- Cluster Sampling
- Multistage Sampling Non-Probability
- Purposive sampling
- Quota Sampling

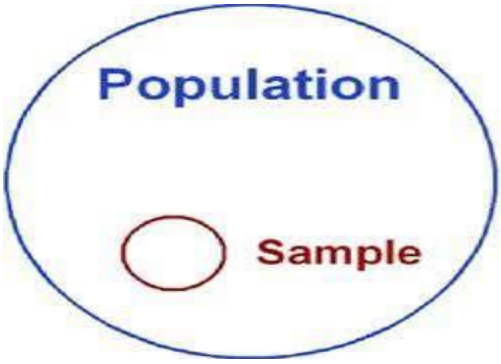


- **Sampling**: It is the process of selecting a representative group from the population. The target population may be as board as all humans but it may be a smaller group as people who are diabetics or only the teenagers.
- **Sample**: A sample is a group of units selected from a population by a defined or simply a part of a statistical population.
- The sample should be representative of the general population. 510% portion of the sample size of total
- **Population**: Population is the total group of individuals from which the sample might be drawn.

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• All members of population possess the same basic and defined characteristics, but varying in amount or quality from one member to another.

Module# 110 Random Sampling

• If every individual of target population has an equal chance of selection then it will be called random sampling. Simple random sampling is the easiest form of probability sampling.

• No biasness and expert opinion is required in this method.

Random Sampling

• All the researcher needs to do is assure that all the members of the population are included in the list and then randomly select the desired number of subjects.

Examples:

• Selection of plants from rice fields

• Selection of students from class

• When carrying out a survey, it would be impractical to study a whole population.

Module# 111 Systematic Sampling

• In this sampling method first sample from the population is taken randomly but after that samples are selected systematically at a regular interval.

Examples:

• Shopping Centre

• Selection of plants

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• Selection of soil sample

Module# 112 Principles of Systematics

Stratified Sampling

Module# 113 Cluster Sampling

• It is a sampling technique where the entire population is divided into groups, or clusters and a random sample of these clusters are selected.

• Members of one cluster have similarity with each other and also with other clusters; minor differences can be there between different clusters.

• Individual selection from each cluster is random.

Cluster Sampling

• Cluster of plants for study

• Cluster of animals

Module# 114 Multistage Sampling

• It is complicated form of cluster sampling. In this type of sampling we divide our population into groups (cluster) and each group is further divided into sub- groups in such a way that each lower group is adjusted into higher group.

Multistage Sampling

• For example in a survey of tuberculosis prevalence in world, if we divide whole world into continents, then select 10countries from each continent randomly, then 10districts from each country, 10 cities from each district, 10 villages from each city and finally 10 families from each villages randomly. This type of END sampling is multistage sampling.

Module# 115 Principles of Systematics

Non-Probability Sampling

• **Quota sampling:** In this type of sampling we divide the population intro different groups or strata based on some specific criterion but selection of individual from each group is not random.

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• **Purposive sampling:** In this technique researcher selects sample based on their knowledge and professional judgment. Purposive sampling can also be known as judgmental, selective or subjective sampling.

For example purchase of car from market based on expert opinion or selection of staff for biology teaching in a college by a panel of subject experts.

Module# 116 Construction of Phonogram

• A tree diagram depicting taxonomic relationships among organisms based on overall similarity of many characteristics without regard to evolutionary history

Rules

•Start from lowest value

•Try to make maximum separate clusters

•Do not repeat

•Incase of same number multiple time, you can chose any combination

• Value of previous similarity should be higher than afterwards

Construction of Phonogram

Construction of Phenogram

Characters	A	B	C	D	E
No. of eyes	8	8	6	6	8
No. of spinnerts pairs	2	2	2	3	3
Shape of sternum	Heart shape	Heart shape	Heart shape	Oval	Oval
Size of Median eyes	Large	Small	Small	small	Large
size of chelicera	Small	small	Large	Large	Large

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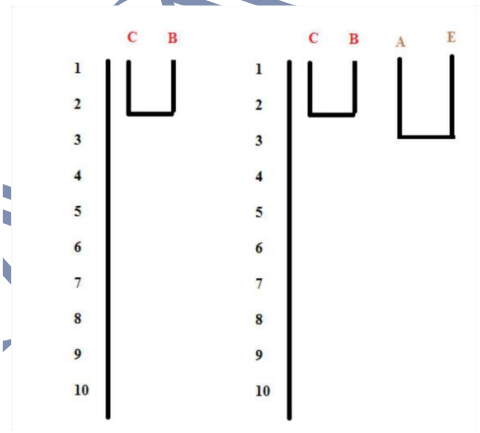
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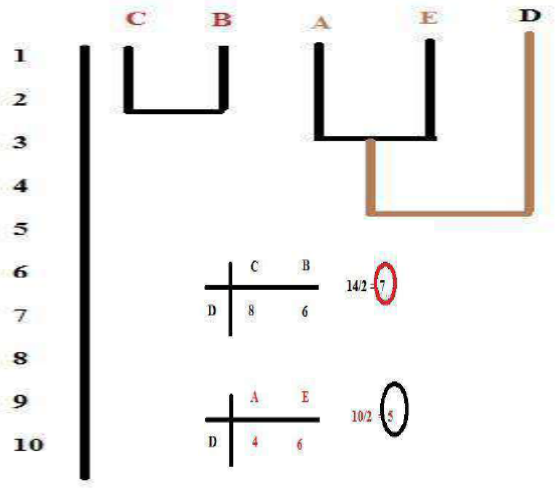
	A	B	C	D	E
A	x				
B	9	x			
C	5	2	x		
D	4	6	8	x	
E	3	5	7	6	x

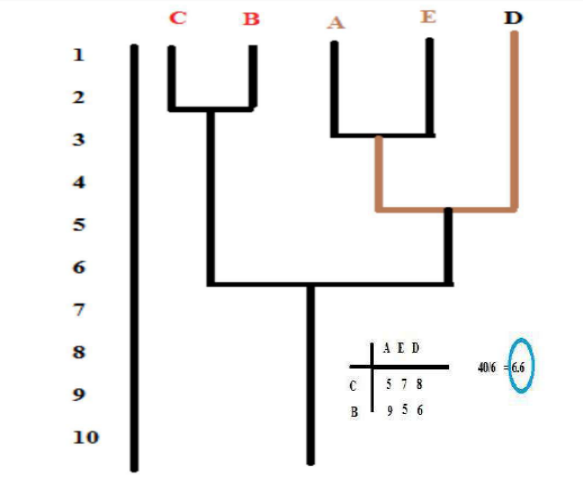
	A	B	C	D	E
A	x				
B	9	x			
C	5	2	x		
D	4	6	8	x	
E	3	5	7	6	x

Module# 117 Construction of Phenogram

	A	B	C	D	E
A	x				
B	9	x			
C	5	2	x		
D	4	6	8	x	
E	3	5	7	6	x





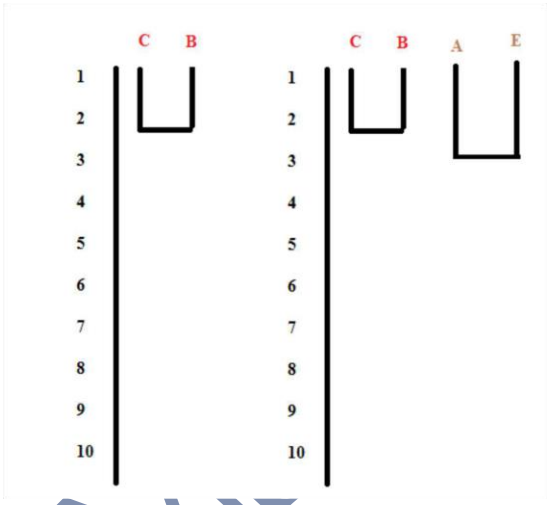


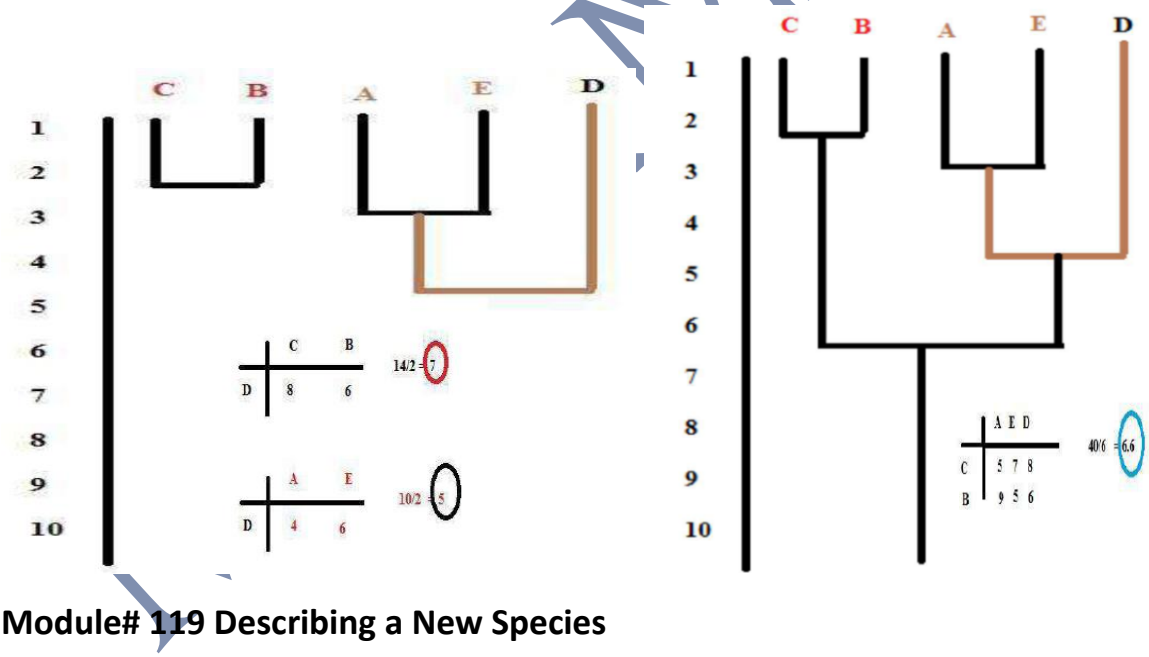
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Module# 118 Construction of Phenogram

	A	B	C	D	E
A	x				
B	9	x			
C	5	2	x		
D	4	6	8	x	
E	3	5	7	6	x





Module# 119 Describing a New Species

•Describing new species is not something that is done by non - taxonomists.

• To describe a new species, you need to have the appropriate credentials, experience, and knowledge to do so.

• Assuming you have all describing a New Species

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• Carrying out the necessary measurements (scales, morphological size, characteristics, etc.) needed in a description.

• Record location, habitat, and other info such as DNA Describing a New Species

• Evidence of comparative studies demonstrating that your species is indeed new or undescribed (this can take a long time to do properly).

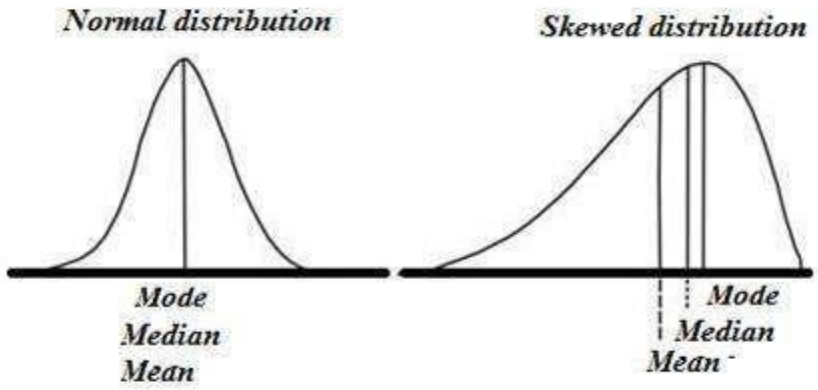
• Finding the right journal for publishing your description.

• Zootaxa, ZooKeys

Module# 120 Normal and Skewed Data

• In normal data values of means, median mode are equal but this not the case for asymmetric or skewed data.

• Normal data follow a bell shape curve. In skewed distribution one tail is larger than other as in above figure.



• In normal data most of the individual values are close to the mean of the data. However, in skewed t the hat difference between individual values and mean values is higher.

• Extreme values are almost negligible in normal data but higher

• Values of standard deviation are less for normal data compared to skewed data.

• Measure of central tendency for normal data is mean but it is median for skewed data.

Module# 121 Parametric Vs Non - Parametric Tests

(VERY IMPORATANT)

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• Parametric tests are used for Normal data

• T – test • Z – test • F – test • ANOVA • Regression • Correlation

• Non - parametric tests for skewed data. • Wilcoxon Signed Rank Test • Mann Whitney U Test

• Kruskal - Wallis test • Principal Component Test

Parametric Test	Non-Parametric test
Make assumptions about the population	Does not make assumptions
Uses normal probabilities	Distribution is arbitrary
Uses mean value for central tendency	Uses median value for central tendency
Require previous knowledge about the population	Does not require previous knowledge about the population

Friedman Tests Parametric Vs Non - Parametric Tests • for measuring the degree of association between two quantitative variables, **Pearson's coefficient of correlation** is used in the parametric test, while **Spearman's rank correlation** is used in the nonparametric test.

Module# 122 Types of data Distribution (VERY IMPORATANT)

• Normal Distribution

• Binomial Distribution

• Poison Distribution

Normal Distribution

• Data which is normally distributed and follow the normal curve is called normal data. It can be continuous and discrete in value.

Binomial Distribution

• Data in which there are only two outcomes from an event and numbers of events are predefined is called binomial distribution.

For example in five test match series, outcomes of each match are two i.e. to win or lose. Outcome of the event is a discrete number.

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Poison Distribution

• The data distribution in which total numbers are not predetermined and some time there number can reach to infinite. These events occur in nature rarely. Outcome of such events is complete discrete value not in decimal points.

For example number of deaths in Lahore per hour. Number of deaths can be infinite if some unusual event occurs.

Module# 123 Measure of Central Tendency and Measure of Variability

Measure of central tendency

• It provide information about average value of a data set.

• Mean, Median and Mode are commonly used measures of central tendency.

• Data set has only one and only one Mean.

• Value of mean is influenced by all observations in the data set.

Measure of variability

• Mean deviation

• Variance

• Standard deviation

• Standard error of mean

• Range

• Less vs high value

Module# 124 Z - test

• Z - test is used to compare the means of two samples when sample size is greater than 30 (n >30).

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$$Z = \frac{\overline{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Z = X (BAR) – U(macro) / SE

• This test like t - test assume a normal distribution of the data.

• This test is most useful when the standard deviation is known.

Z - Test

• Test the hypothesis that there is no difference in the calculated sample mean with given population means.

$$z = \frac{\overline{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

• Population mean: μ = 100, n= 55, Sample mean: and = 12

• By putting the values in formula:

• Z - calculated= - 2.7

Z - Test

• Critical value at 95% confidence interval for Z - test is ±1.96. Z Calculated is not within the acceptable range so our null hypothesis is rejected.

Module# 125 ANOVA

• Analysis of variance (ANOVA) is a technique used to compare means of more than two samples.

• This technique can be used only for numerical data as it is a parametric test."

Assumptions

• Number of samples must be more than two.

• Data should be normally distributed.

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• Samples must be independent.

• Homogeneity of variances.

Examples

• A group of psychiatric patients are trying three different therapies: counseling, medication and biofeedback. You want to see if one therapy is better than the others.

• Students from different colleges take the same exam. You want to see if one college outperforms the other.

Module# 126 ANOVA

Source of variation	SS	DF	MS	F-value
Between group variation				
Within group variation				
Total				

• SS (total) [Sum of square] = (Each observation – Grand mean) 2(whole Square)

• SS (within) = ((Each observation – Group mean) 2 (whole Square)

• (between) SS = (Group mean Grand mean) 2 (whole Square) × Number of observation in each group

• DF (between) = Total number of Groups – 1

• DF (within group) = Total number of Observations – Total number of Groups

Module# 127 One - Way ANOVA

• A one - way ANOVA uses one independent variable, while a Two - way ANOVA uses two independent variables.

• As a crop researcher, you want to test the effect of three different fertilizer mixtures on crop yield. You can use a one - way ANOVA to find out if there is a difference in crop yields between the three groups.

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• You are interested to compare the effectiveness of three medicines in treating cancer patients

• Comparison of four feeds on weight of fish

Replicates	Lahore	Sialkot	Sargodha
1	8	10	8
2	14	14	6
3	3	11	5
4	7	9	1
Mean	**8**	**11**	**5**

Grand Mean = 8+11+5/3 = 8

• SS (total) [Sum of square] = (Each observation – Grand mean) 2 (whole Square)

• SS (Total) = (8 - 8) 2 (whole Square) + (14 - 8) 2 (whole Square) + (38) 2 (whole Square) + (7 - 8) 2 (whole Square) + (10 - 8) 2 (whole Square) + (14 - 8) 2 (whole Square) + (118) 2 (whole Square) + (9 - 8) 2 (whole Square) + (8 - 8) 2 + (6 - 8) 2 (whole Square) + (6 - 8) 2 + (5 - 8) 2 (whole Square) + (1 - 8) 2 (whole Square)

• **SS (Total) = 174**

SS (within) = ((Each observation – Group mean) 2

• SS(within) = (8 - 8) 2 +(14 - 8) 2 + (38) 2 +(7 - 8) 2 +(10 - 11) 2 +(14 - 11) 2 +(11 - 11) 2 +(9 - 11) 2 +(8 - 5) 2 +(6 - 5) 2 +(5 - 5) 2 +(1 - 5) 2

•**SS (with in) = 102**

Module# 128 One - Way ANOVA

• **SS (between)** = (Group mean Grand mean) 2 × Number of observation in each group

• SS (between) = [(8 - 8) 2 × 4] + [(118) 2 × 4] + [(5 - 8) 2 × 4] = 72

• **DF (between)** = Total number of groups – 1

• DF (between) = 3 - 2 1 =

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Source of variation	SS	DF	MS	F-value
Between group variation	72	2	36	3.17
Within group variation	102	9	11.33	
Total	174	9+2 = 11	-	

•**DF (within group)** = Total number of observation – Total number of groups

• DF (within group) = 12 - 3 = 9

MS (between)

• It can be calculated by dividing the

SS (between groups) with DF (between groups)

• 72 /2 = 36

MS (within)

It can be calculated by dividing the SS (within group) with DF (within group)

• 102/9 =11.33

F - Calculated

•3.11

F -Tabulated

• Comparison of both

• Decision about hypothesis

Module# 129 Two - Way ANOVA

• In two - way ANOVA there are two independent variables and one dependent variable.

• In one - way ANOVA there was only one independent variable

Two - Way ANOVA Assumptions

• The population must be close to a normal distribution.

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• Samples must be independent.

• Population variances must be equal.

• Groups must have equal sample sizes.

• One could use a two - way ANOVA to understand whether there is an interaction between gender and educational level on test anxiety amongst university students, where gender (males/females) and education level (undergraduate/postgraduate) are your independent variables, and test anoxia ty is your dependent

• For example, a two - way ANOVA allows a company to compare worker productivity based on two independent variables, such as department and gender.

• It is utilized to observe the interaction between the two factors. It tests the effect of two factors at the same time.

• The effective life (in hours) of batteries is compared by **material** type (1, 2 or 3) and operating **temperature**: Low (- 10 ˚C), Medium (20˚C) or High (45˚C). Two - Way ANOVA

• Twelve batteries are randomly selected from each material type and are then randomly allocated to each temperature level.

Module# 130 T - test

• T - test is appropriate whenever you want to compare the means of two samples.

• This test is used to find out whether the means of two samples are statistically different from each other or not.

• Data should be normally distributed.

• Samples should be taken randomly from the population

• Sample size should be 30 or less than 30.

• All randomly collected samples should have the same variances (within the limits of sampling error).

• Standard deviation may or may not be known.

• Independent t - test/two sample t-test

• Paired t – test

Module# 131 Independent t - test

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•For two sample t - test, the two populations must be independent

• It is not necessary that sample size should be same for both samples.

• Examples: Test the hypothesis that means heights (cm) of plants randomly selected from two wheat fields are same.

Data of field I (X_1)= 10, 15, 20. 25 and 30.

Data of field II (X_2)= 4, 8, 10. 15 and 10.

Null hypothesis: $\bar{X}_1 = \bar{X}_2$

Alternative hypothesis is $\bar{X}_1 \neq \bar{X}_2$.

Formula: $$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

X_1	X_2	X_1^2	X_2^2
10	4	100	16
15	8	225	64
20	10	400	100
25	15	625	225
30	10	900	100
$\sum X_1 = 100$	$\sum X_2 = 47$	$\sum X_1^2 = 2250$	$\sum X_2^2 = 505$

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$$\sigma_1^2 = \frac{\Sigma(X_1 - \bar{X}_1)^2}{n-1}$$

$$= \frac{\Sigma X_1^2 - \frac{(\Sigma X_1)^2}{n}}{4} = \frac{2250-2000}{4} = 62.5$$

$$\sigma_2^2 = \frac{\Sigma(X_2 - \bar{X}_2)^2}{n-1}$$

$$= \frac{505 - \frac{(47)^2}{5}}{4} = \frac{505-441.8}{4} = \frac{63.2}{4} = 15.8$$

$$t = \frac{20-9.4}{\sqrt{\frac{62.5}{5} + \frac{15.8}{5}}}$$

$$= \frac{10.6}{\sqrt{12.5+3.6}} = \frac{10.6}{\sqrt{15.66}} = \frac{10.6}{3.95} = 2.68$$

$$\text{Null hypothesis: } \bar{X}_1 = \bar{X}_2$$

t = 2.68 (calculated value)

As both samples are independent so degree of freedom will be n-1 (from sample X_1) and n-1 (from sample X_2).

5-1 = 4 (X_1) and 5-1 = 4 (X_2)

Degree of freedom: 4+4 =8

Tabulated value/standard value/critical value = 2.3

Result: As our calculated value is lower than tabulated value so our null hypothesis is accepted.

Module# 132 Paired t - test

X	Y	d	d^2
9	12	-3	9
12	14	-2	4
13	19	-6	36
20	25	-5	25
5	10	-5	25
		$\sum d = -21$	$\sum d^2 = 99$

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$$t = \frac{\bar{d}}{\sqrt{\frac{\sigma^2 d}{n}}}$$

$$\sigma^2 d = \frac{\sum(d - \bar{d})^2}{n - 1}$$

$$= \frac{\sum d^2 - (\bar{d})^2}{n-1}$$

$$= \frac{99 - 17.64}{4} = \frac{81.36}{4} = 20.34$$

$$\bar{d} = \frac{-21}{5} = -4.2$$

$$t = \frac{-4.2}{\sqrt{\frac{20.34}{5}}} = \frac{-4.2}{\sqrt{4.068}} = \frac{-4.2}{2.016} = -2.08$$

T calculated: -2.08

T tabulated $=2.77$

As observation were paired so degree of freedom (n-1)= 4.

Our null hypothesis is rejected as T- calculated is smaller than T-tabulated.

Module# 133 Standard Deviation

- It is positive square root of variance.
- Standard deviation "provides a sort of average of the differences of all scores from the mean.
- It is a measure of the dispersion of scores around the mean.

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$$SD = \sqrt{\frac{\sum(x - \bar{x})^2}{n - 1}}$$

• Generally, a low standard deviation means that a set of scores is not very widely dispersed around the mean.

• While a high standard deviation indicates that the scores are more widely dispersed

• Its advantage over variance is that its physical unit are same as that of individual observations

• The standard error is also a measure of spread.

• It is represented by SE.

$$SE = \frac{\sigma}{\sqrt{n}} \quad \text{or} \quad SE = \frac{SD}{\sqrt{n}}$$

• Higher value of SE indicates higher spread of the data.

• The difference in the standard error and standard deviation is that standard error uses sample data but standard deviations use population data.

• SE tells us that how far our sample mean deviate from Standard Deviation population means.

Module# 134 Standard Deviation

• The standard deviation is a measure of the spread of scores within a set of data.

• Usually, we are interested in the standard deviation of a population.

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The **sample standard deviation formula** is:

$$s = \sqrt{\frac{\sum(X - \bar{X})^2}{n-1}}$$

where,

s = sample standard deviation
$\sum$ = sum of...
$\bar{X}$ = sample mean
n = number of scores in sample.

• However, as we are often presented with data from a sample only, we can estimate the population standard deviation from a sample standard deviation.

• The standard deviation is a measure of the spread of scores within a set of data.

• A **sample standard deviation** is a statistic. This means that it is calculated from only some of the individuals in a population.

• The **population standard deviation** is a parameter, which is a fixed value calculated from every individual in the population.

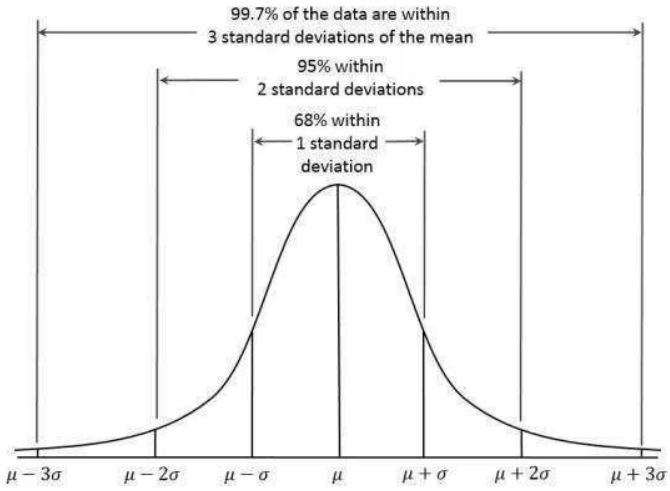
Relationship between Variance and SD

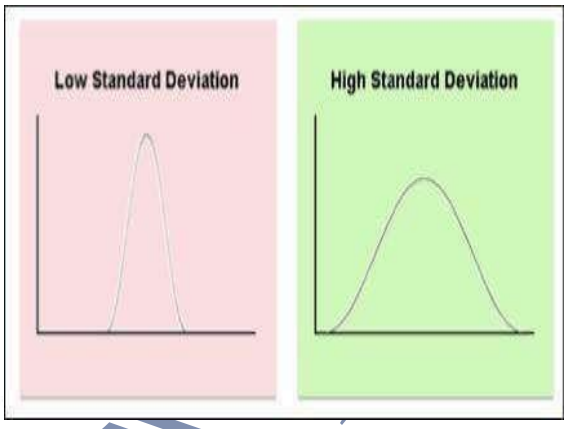
• The standard deviation is the square root of the variance. The standard deviation is expressed in the same units as the mean is, whereas the variance is expressed in squared units,

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Module# 135 Introduction to Spiders

- Most abundant natural predators (Maloney et al., 2003)
- Ranked 7th in animal diversity (WSC, 2018)
- More than 46000 species
- Strictly carnivorous
- Cause wasteful killing
- Show functional and numerical response
- Best survivors under food shortage
- Prey switching
- Cannibalistic
- Top down effect
- Indirect killing

Module# 136 Introduction to Spiders

- Ground runners
- Stalkers
- Web builders
- Bio - control agents
- Model organism for research
- Spiders silk

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• Spider venom peptide

Module# 137 Morphology of spider Leg

• Spiders typically have eight walking legs (insects have six)

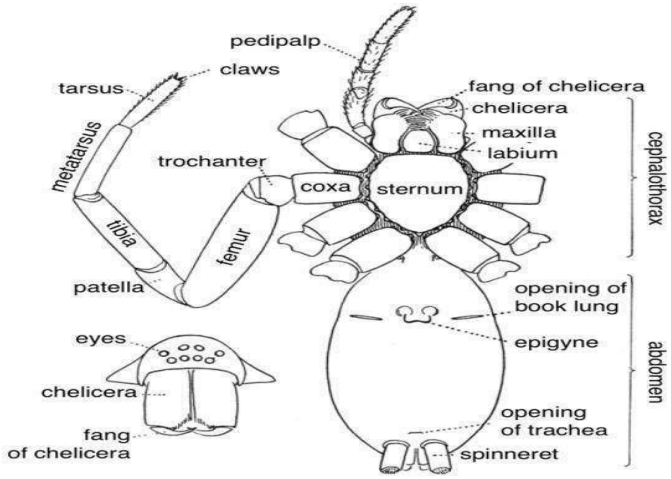
• They do not have antennae

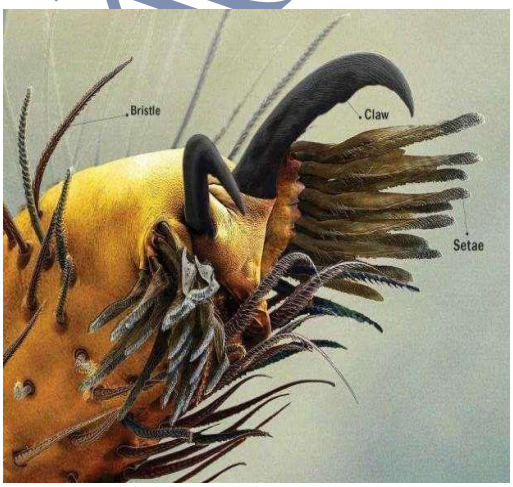
• The pair of appendages in front of the legs are the Pedi palps (or just palps)

• Spiders' legs are made up of seven segments

• The tip of the tarsus bears claws, which vary in number and size

• Spiders that spin webs typically have three claws, the middle one being small





• Hunting spiders typically have only two claws

• Starting from the body end, these are the: • Coxa • Trochanter • Femur • Patella • Tibia

• Metatarsus • Tarsus.

• Since they do not have antennae, spiders use specialized and **sensitive setae** on their legs to pick up scent, sounds, vibrations and air currents.

• Some spiders, such as not have claws. The Australia crab spiders, do not have claws.

• The pedipalps have only six segments: the metatarsus is missing.

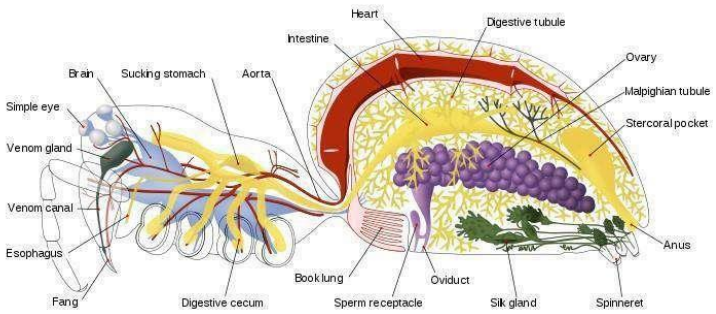
• In adult males, the tarsus of each palp is modified for mating (variously called a palpal bulb, palpal organ or copulatory bulb)

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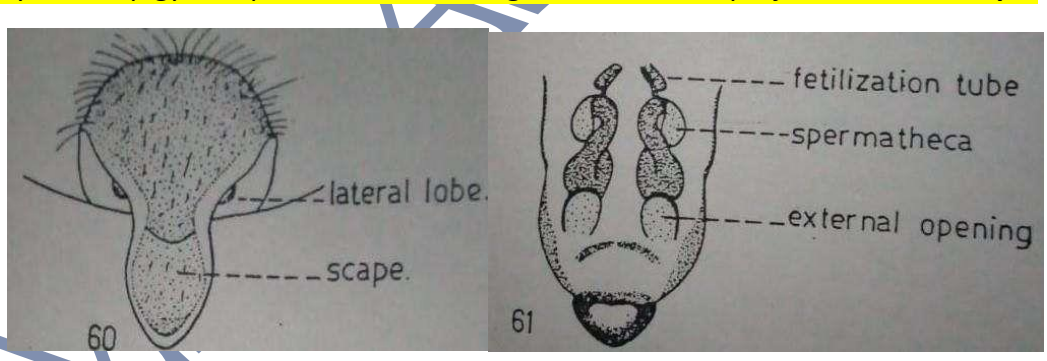
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Module# 138 Morphology of spider's Abdomen

- The abdomen is also known as the **opisthosoma**.
- On the ventral side of the abdomen are two hardened plates covering the book lungs.
- Females genital organ is also present on ventral side of the abdomen.



- It is called **epigyne**.
- These are called the **epigastric plates**
- In many cases epigyne is provided with a tongue like sclerotic projection called **scape**.



- Scape often have paired lateral lobes as in Neoscon Tikadar, 1982. The Funa of India
- Epigine has two external openings which lead to sperm thecae inside and paired of narrow **curved tubes called fertilization tube.**
- Scape often have paired lateral Tikadar, 1982. The Funa of India lobes as in Neoscon
- A fold, known as the epigastric furrow, separates the region of the book lungs and epigyne.

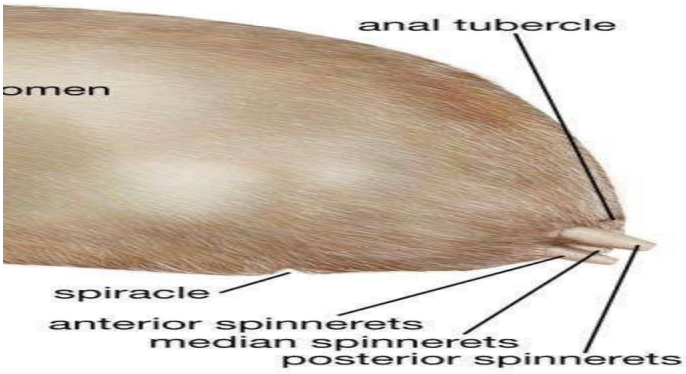
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• In the middle of this furrow is the opening of the oviduct (in females) and at either end are the lung slits.

• On posterior side of the abdomen there is conical anal tubercle that bear three pairs of spinnerets ventrally.

• On ventral side there are two pairs of book lungs.



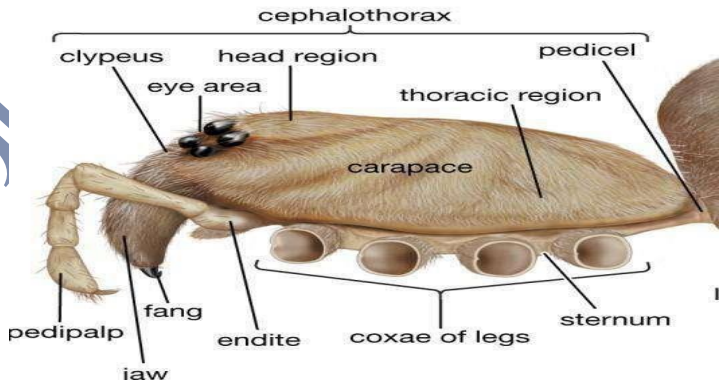
Module# 139 Morphology of spider's Cephalothorax

• Cephalic region has 6 - 8 eyes

• Eyes or black or diurnal

• White or nocturnal

• If only one type of eyes (homogenous)



• If both type of eyes or present (heterogeneous) usually arranged in two rows

• Each row usually contain 4 eyes. • Rows of eyes are frequently curved.

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• Anterior median/Anterior lateral • Posterior median/Posterior lateral

• Cephalic area of eyes is called **ocular area.**

• Area of four median eyes is called **ocular quad.**

• The area between anterior row of eyes and anterior margin of carapace is **clypeus.**

• The depression in the middle of thorax is called **thoracic groove or fovea**.

• Appendages • Chelicerae/Fangs

• Padipalp (2nd pair of appendages) • Papl has six parts/metatarsus missing.

• In males tarsus/modified copulatory organ • Ventral side is sternum

• Oval/heart shape

Module# 140 Morphology of spider's Spinnerets

• The abdomen of spiders has modified pairs of movable telescoping organs called **spinnerets,** which produce silk.

• The suborder **Mesothelium** is unique in having only two types of silk glands – thought to be the ancestral condition.

• Depending on the complexity of the species, spiders will have two to eight spinnerets, usually in pairs.

• Behind each spinneret visible on the surface of the spider lies a gland.

• Each spigot has a valve that controls the thickness and speed of the extruded material. As the spigots pull the **fibroin protein molecules** out of the ducts and extrude them into the air, the molecules are stretched out and linked together to form long strands.

• The **spinnerets** wind these strands together to form the sturdy silk fiber.

• Most spiders have multiple silk glands, which secrete different types of silk material optimized for different purposes.

• Spider silk is incredibly strong and flexible. Five times as strong as an equal mass of **steel** and twice as strong as an equal mass of **Kevlar**.

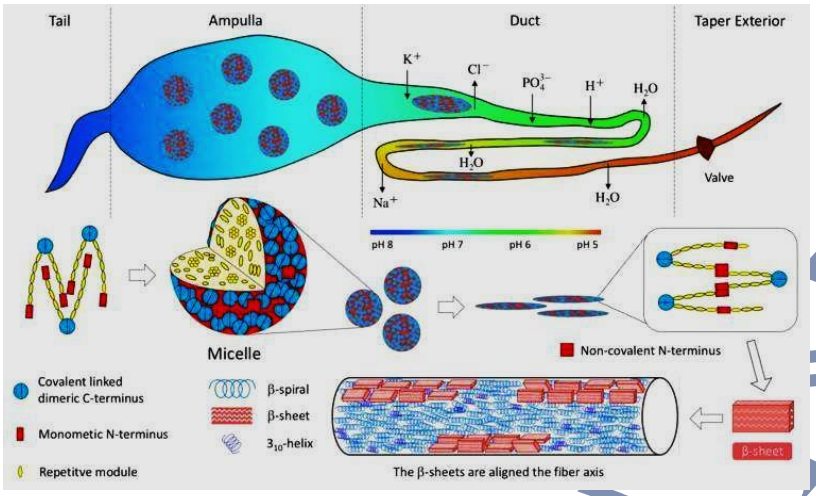
• Elasticity

• Size Temperature/PH/uses

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Module# 141 Introduction to Scorpions

• Scorpions are predatory arachnids of the order Scorpions.



• They have eight legs, and are easily recognized by a pair of grasping pincers and a narrow, segmented tail, often carried in a characteristic forward curve over the back and ending with

• Scorpions primarily prey on insects and other invertebrates, but some species take vertebrates.

• They use their pincers to restrain and kill prey.

• The vast majority of species do not represent a serious threat to humans, and healthy adults usually do not need medical treatment after being stung.

• Poisonous scorpions • Habitat/desert/Mountains • Collection/UV/Turning stone/pitfalls

Module# 142 Introduction to Scorpions

• Scorpion bites are highly painful

• Venom peptides

• Type/Long chain/Short chain

• Medical importance/Buthidae

• Treatment of diseases

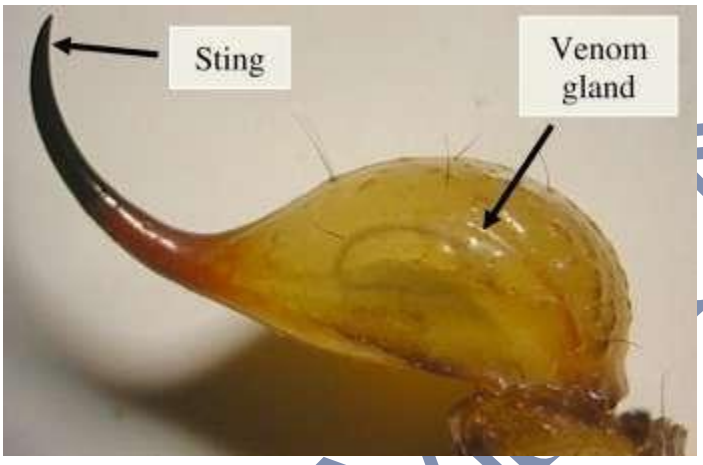
• Cancer treatment

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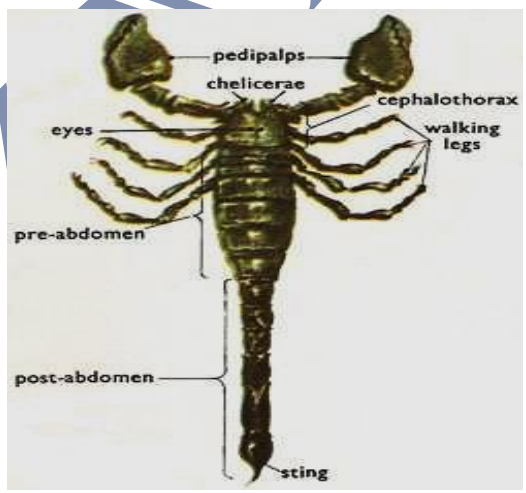
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- Venom price
- Venom collection
- How to collect venom
- Effect of temperature on venom production
- Amount of venom
- Method of venom collection



Module# 143 Morphology of Scorpions

• Cephalothorax • Padipal • Chelicera • Abdomen • Walking legs • Sting



• Body is covered with a cuticle consisting of chitin and tanned proteins • 8.5-23mm

• The opisthosoma is subdivided into a broad anterior portion, the mesosoma or pre-abdomen, and a narrow tail-like posterior, the metasoma or post-abdomen.

• Most species are nocturnal. • Feeding habit/cannibalism

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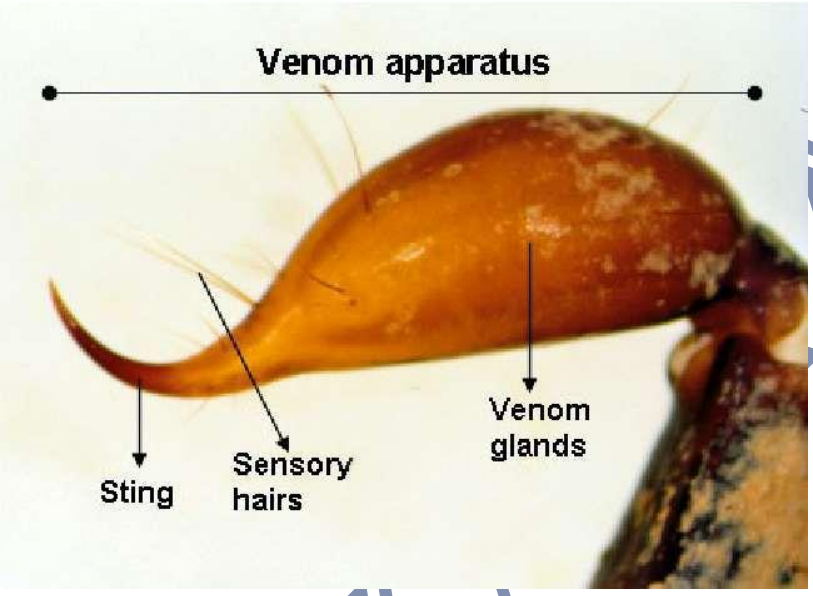
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• The telson includes the vesicle, which contains a symmetrical pair of venom glands.

• Externally it bears the curved stinger, the hypodermic aculeus, equipped with sensory hairs.

• Abdominal segments these segments have keels, setae and bristles which may be used for

• Each of the venom glands has its own duct to convey its secretion
along the aculeus from the bulb of the gland to immediately sub terminal of the point of the aculeus.

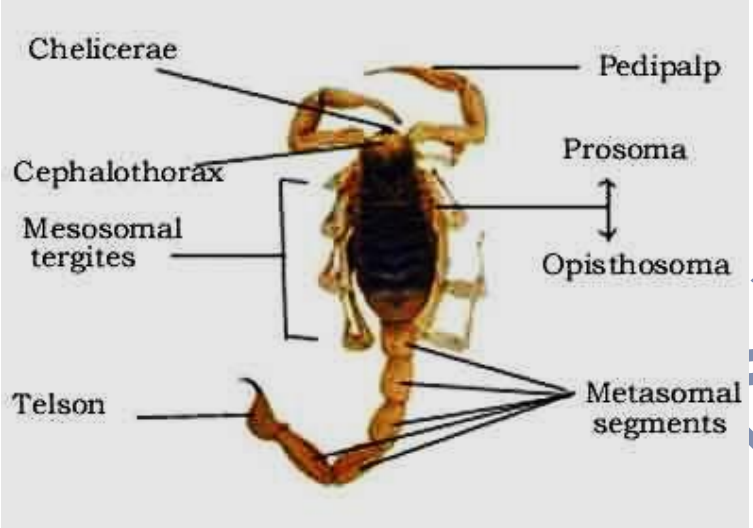


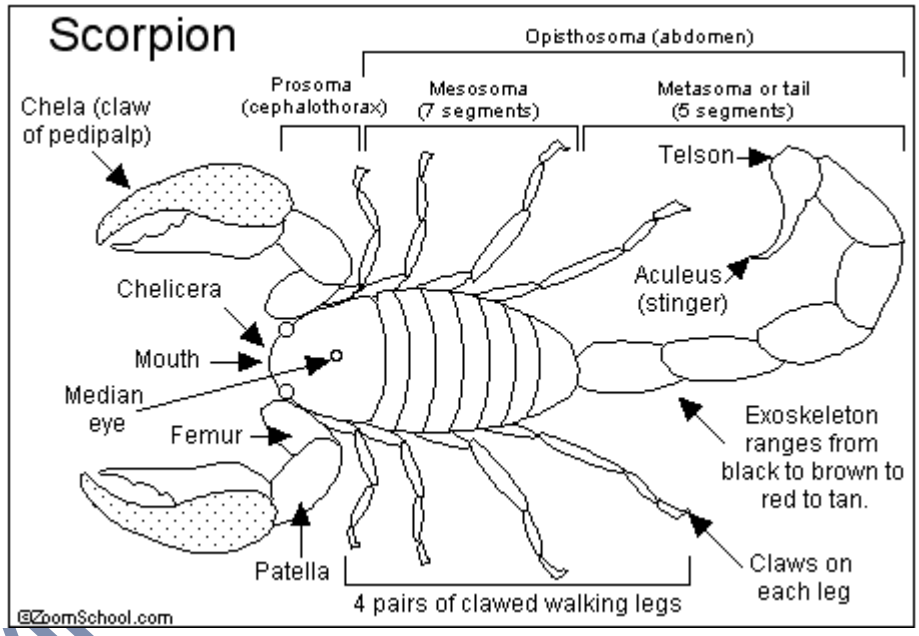
• Parts of scorpion body

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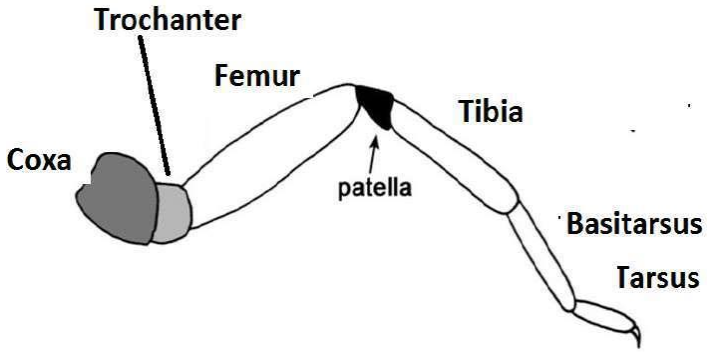


Module# 144 Morphology of Scorpion Leg

• Four pairs of legs • each leg is terminating in a pair of claws.

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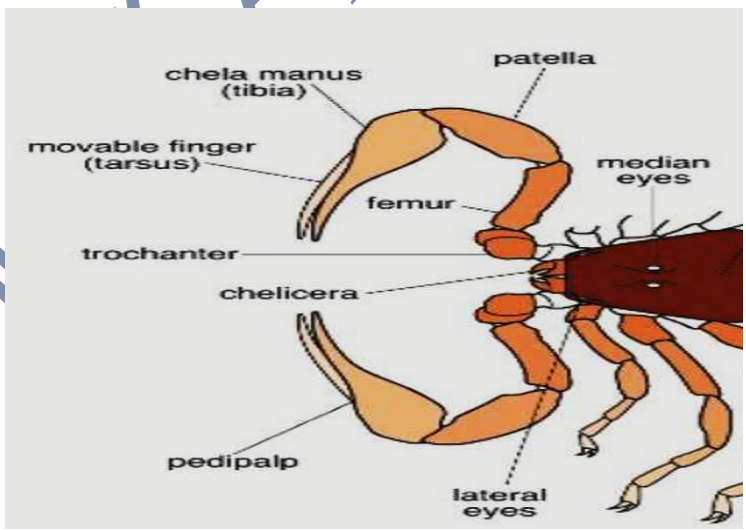
• A series of trichobothria, the number and positions of which are taxonomically informative, is present on the pedipalp femur, patella, and tibia.

• Each chelicera comprises three segments, including a chela manus with fixed finger and a movable finger.

• Each pedipalp comprises a coxa, trochanter, femur, patella, chela manus with fixed finger (tibia), and movable finger (tarsus).

• Setae are thick hairs which cover the legs, body, and pedipalps (leg-like mouthparts).

• They are sensitive to the touch.



• Also located on the pedipalps are smaller hairs known as trichobothria, which sense air vibrations.

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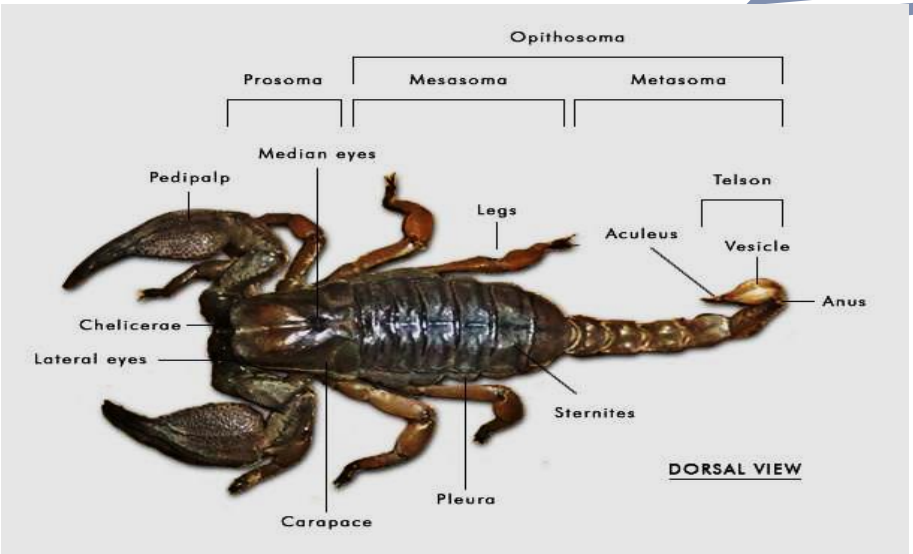
• Ground vibrations are sensed by other organs located on the tip of the legs.

Module# 145 Morphology of Scorpion Telson

• The telson is present in Metasoma.

• Metasoma is also called as post- abdomen.

• The post abdomen includes five metasomal segments, the last of which contains the anus and bears the telson.



• The telson consists of the vesicle, which holds a pair of venom glands, and the hypodermic aculeus, the venom-injecting sting.

• Each of the venom glands has its own duct to convey its secretion along the aculeus from the bulb of the gland to immediately sub terminal of the point of the aculeus, where each of the paired ducts has its own venom pore

• **An extrinsic muscle system** in the tail moves it forward and propels and penetrates with the aculeus, while an **intrinsic muscle system** attached to the glands pumps venom

• All scorpion species possess venom, which is used to kill or paralyze prey and as a defense against predators.

Module# 146 Morphology of Chelicerae and Abdomen

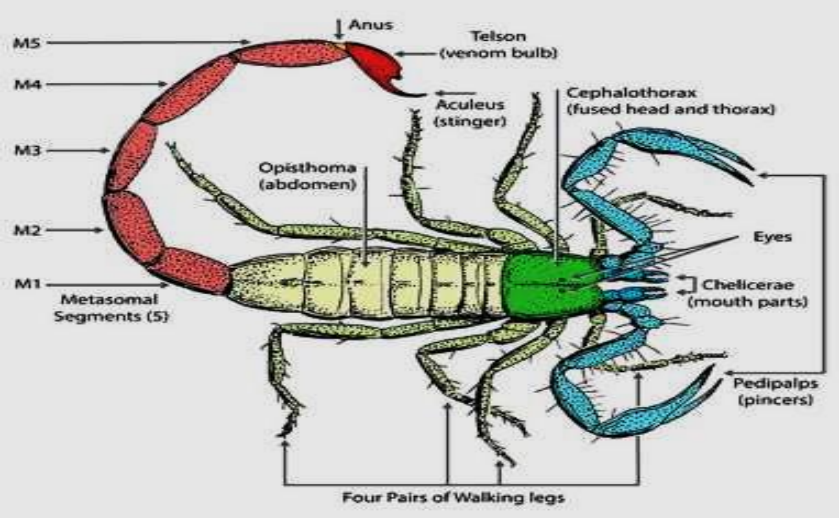
• The chelicerae of scorpion is present in cephalothorax region.

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• They are scorpion's jaws.

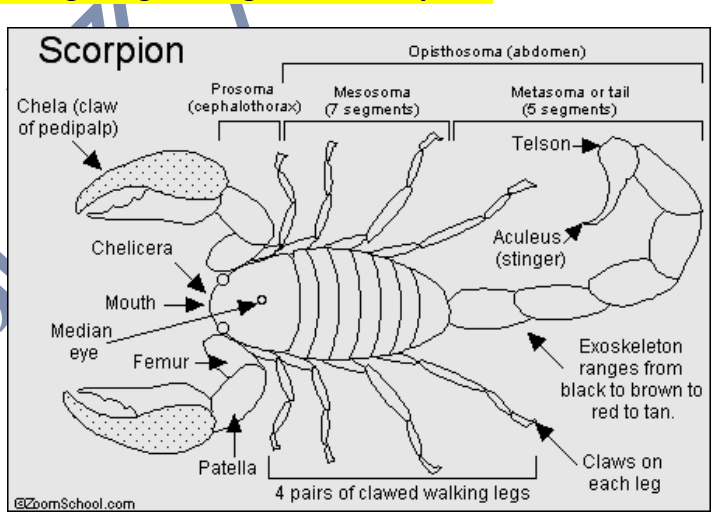
• Each chelicera comprises three parts a coxa, tibia which has a fixed finger, and the tarsus forming a moveable finger.



• Within the abdomen are the scorpion's respiratory system, which is a set of thin plates called **book lungs**.

• A spiracle leads to each pair of book lungs on each segment.

• The opisthosoma has eight legs arranged into **four pairs**.



Module# 147 Accumulation Curve

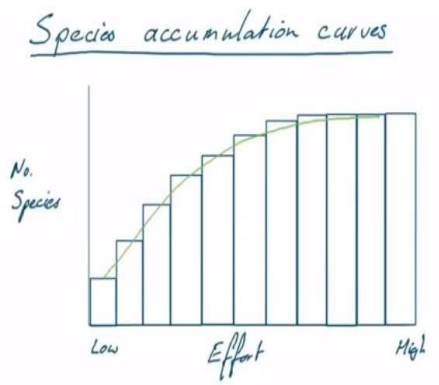
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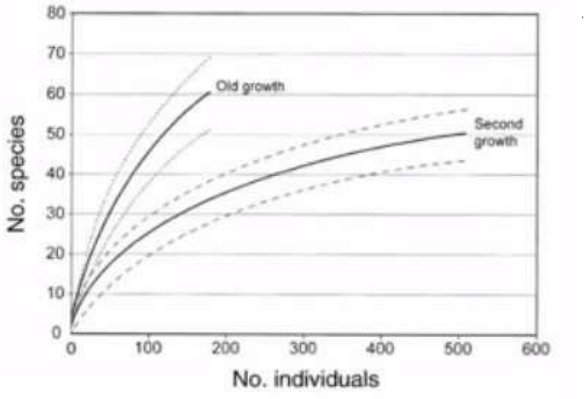
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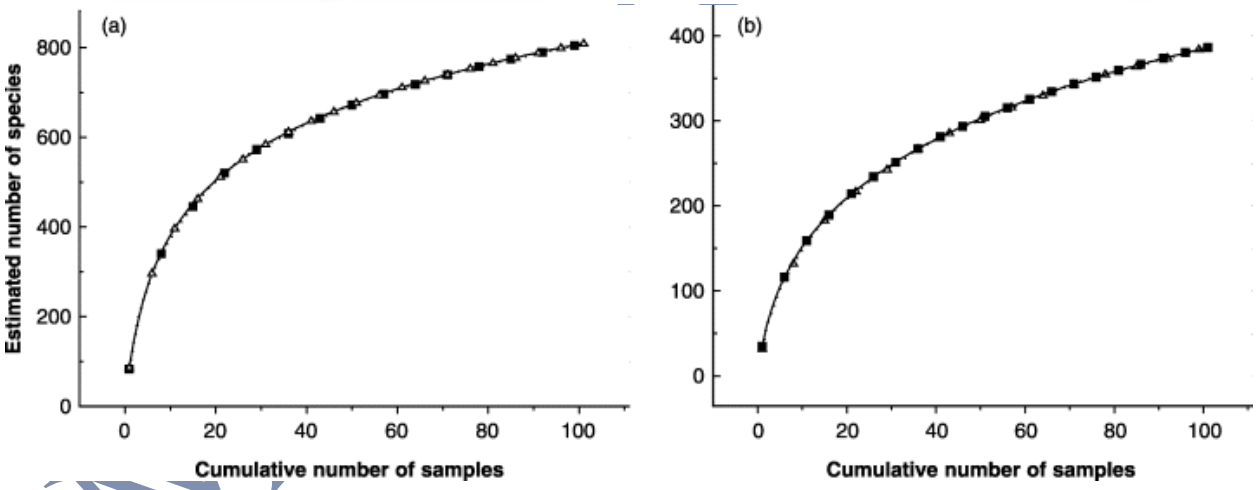
• The species discovery curve or species accumulation curve is a graph recording the cumulative number of species of living things recorded in a particular environment as a function of the cumulative effort expended searching for them.

• Types of accumulation curve

• Importance of accumulation curve
• Accumulation curve vs trapping effort
• Accumulation curve vs trapping method
• Accumulation curve vs biasness







Module# 148 Calculation of Allelic and Genotypic Frequency

Hardy-Weinberg Equilibrium Model

• P = frequency of Allele A
• q = Frequency of Allele a
• P2 = Frequency of AA genotype
• 2pq= Frequency of Aa genotype
• q2 = Frequency of aa genotype

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$$p + q = 1$$

$$(p + q)^2 = 1$$

$$p^2 + 2pq + q^2 = 1$$

	AA	Aa	aa	
Individuals	70	80	50	(200)
Genotypic Array	70/200 = 0.35	80/200 = .40	50/200 = 0.25	= 1
"A' alleles	140	80	0	220/400 = 0.55
'a' alleles	0	80	100	180/400 = 0.45

Module# 149 Calculation of Allelic and Genotypic Frequency

Hardy-Weinberg Equilibrium Model

• In a population there are 320AA individuals, 160 heterozygous (Aa) and 20aa individuals.

• Calculate the genotypic and allelic frequencies.

ALLELIC FREQUENCIES

A 320+320+160/1000 =0.8

a 160+40/1000 =0.2

Alleles = A, a

Genotypes = AA, Aa, aa

Frequency of allele A: $f(A) = f(AA) + 1/2\, f(Aa)$

Frequency of allele a: $f(a) = f(aa) + 1/2\, f(Aa)$

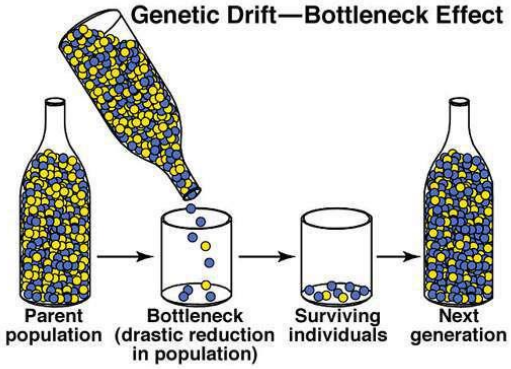
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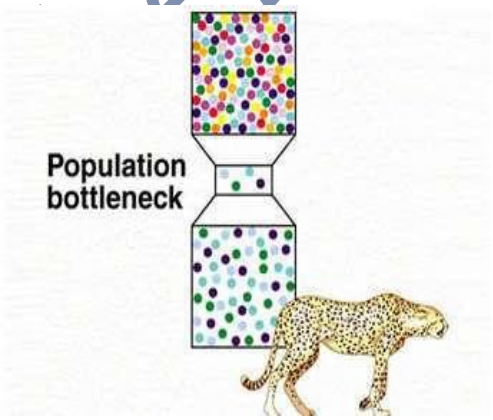
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Genotypes	AA	Aa	aa
Number of individuals	320	160	20
Genotypic frequencies	320/500=0.64	16/500=0.32	20/500=0.04

Module# 150 Genetic Drift and its impact on Allele Frequency

- Change in allele frequency which occur by chance.
- Loss in alleles
- Loss in genetic diversity
- Examples
- Lost genes forever
- Bottle neck effect
- Unique genes





Module# 151 Principles of Systematics

Concept of fitness

- Survival+ reproductive success
- Relative fitness
- Ideal population
- Maximum value of fitness
- Genetic load/Genetic burden
- Population with high load
- Population with low load
- Variability in fitness

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Genetic Load = Max. fit individual - Ave. fitness of population

Genetic Load = 9-6 =3

Module# 152 Taxonomic publications

• No taxonomic study is complete until its results are published.

• Advances in any area of science is made through publications.

• Taxonomic publications range from short description of species (may be one page) to monograph or handbook (several volumes)

✓ Short Research Paper:

They are ordinary descriptive papers of new species, subspecies or combined all of these or some.

• May or may not contain key for identification.

• May contain a brief check list.

✓ Synopsis & Reviews:

They are brief summaries of current knowledge of a group.

• Not necessary to include new material or new interpretations.

• Such work bring scattered work together

• Helpful for Revisionary or monographic work.

Module# 153 Taxonomic publications

✓ Revisions

Revisions are representations of new material or new interpretations that have been integr ated with previous knowledge with brief summaries i.e.

• Restudy of complete unit of particular group of Taxa.

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• Revisions usually include super families, families or part of families.

• It can also deal with genus or species.

• Generic revision

• Species revision

•It include re_description or diagnosis of little known taxa, list of synonymy, new names along with keys.

- ✓ **Monographs**:

A comprehensive systematic study of a particular taxonomic group.

• Always taxon-based, not geography-based.

• Its preparations require more complete material than other types of publications.

• 3-5 years for completion. • Monographs try to review all the facts about a taxonomic group.

•A Monograph is a complete global account of a taxon of any rank – family, genus or species at a given time.

• Book format

- ✓ **Atlases:**

They show comprehensive characters of animals in picture form.

• Purpose of atlases are purely taxonomic

• Very helpful taxonomist for identification

• Comparison of species

Module# 154 Taxonomic publications

- ✓ **Faunal/Floral Work**

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It deals with detail description faunal/Floral work of particular region.

• It may include information on geographical distribution.

• Also include quantitative data and ecological comments.

• Only experience specialists.

• Example: Altas of scale insects of North America

✓ Field Guides/Manuals

• Some works are done by taxonomists for non-taxonomists to help them identify common animals.

• Include easy and understandable simple key characters.

• Manuals are also written in simple language.

• For students and layman

✓ Catalogues & Check lists

A catalogue is an index to taxa arranged in such a way to give a vivid picture of references for both zoological and nomenclature purpose:

Catalogue contain following information:

• The original description reference

• Later references

• Type locality

• Depository detail

• Type of genus

• Taxa are written alphabetically

• Merits depends on completeness

• Its preparation is tedious jobe

•Useful to know many species/genera are reported from particular area.

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Module# 155 Parts of thesis & Research paper

- ✓ **Thesis**

- Abstract/Summary
- Introduction
- Review of literature
- Materials and Methods
- Results
- Discussion
- References

- ✓ **Research paper**

- Abstract
- Introduction
- Materials and Methods
- Results
- Discussion
- Acknowledgements
- References

Module# 156 Parts of thesis & Research paper

- ✓ **Parts of references**

- Name of authors
- Year of publication
- Title/topic of article
- Name of Journal
- Volume number
- Issue number
- Page number

Reference in Text

- Spider is carnivorous animal (Sherawat et al., 2009a)
- Spider is carnivorous animal (Sherawat et al., 2009b)

Module# 157 How to write References

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Full reference

- Serawat, S.M., Butt, A. and Tahir, H.M., 2009. Effect of pesticides on agrobiont spide rs in laboratory and field. Pakistan Journal of Zoology, 42 (4): 120129.

Reference

- Tahir, H.M., 2009. Effect of pesticides on agrobiont spiders in laboratory and field. Thesis submitted to University of the Punjab, Lahore, Pakistan.

Reference of Thesis

- Tahir, H.M., 2009. Effect of pesticides on agrobiont spiders in laboratory and field. Thesis submitted to University of the Punjab, Lahore, Pakistan.

Reference of Book

- Tahir, H.M., 2009. Effect of pesticides on agrobiont spiders in laboratory and field. Oxford University press, England. Pages. 120-129.

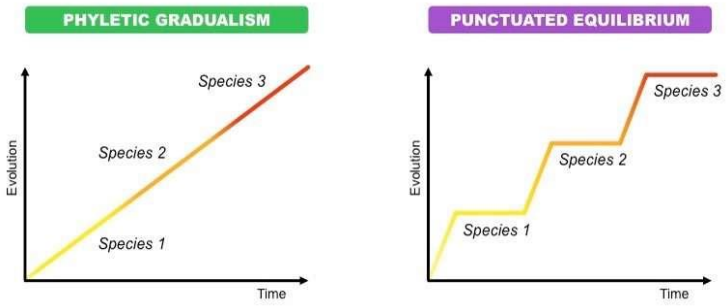
Module# 158 Phyletic Gradualism Vs Punctuated Equilibrium

- ✓ **Phyletic gradualism**

• Evolution is a continuous process

• Bring small variation to particular species

• Occurred over a long period of time



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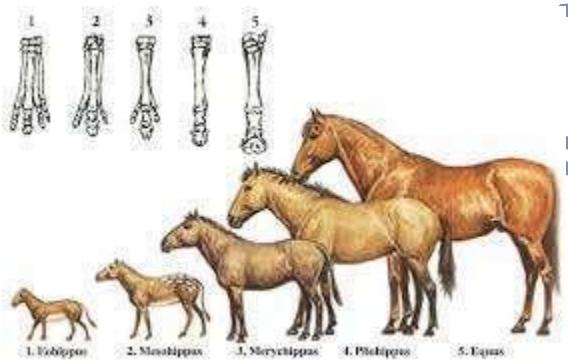
• Describe as slow process

• Not proven by fossil record

• It is natural theory of natural selection.

• Small changes are pased from one generation to other generation.

• Horses 50 million years (from small to model day horse)



✓ **Punctuated Equilibrium**

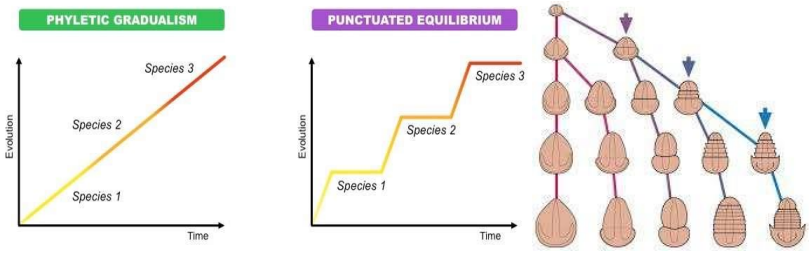
• Evolution is not a continuous process/fast at one time and slow at other time

• Caused consistent and accumulative changes

• Occurred within short period of time

• Better model to describe species

• Evolution of Trilobites



Module# 159 Adaptation

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Adaptation refers to all the structural, functional and behavioral characteristics that enhance the organism's reproductive success in its natural environment.

Adaptation Examples

- **Frogs**: Have adapted to freezing temperatures by evolving a glycol like substance in t heir bloodstreams - like antifreeze.
- **Bats**: Have adapted their hearing to hunt at night via sonic response.
- **Butterflies**: non toxic ones have adapted to look like their more poisonous cousins.
- A **behavioral adaptation** in the oceans is the use of loud, low frequency calls by fin whale to communicate with other whales over great distancs.
- **One physical adaptation** used in the intertidal zones is a crab's hard shell, which protects it from predators, drying out and being crushed by waves.

Module# 160 Taxonomy and Ecology of spiders

(Every LINE is important in These Next lectures)

Spiders are poisonous, predacious and fluid feeding carnivorous animals.

• They are found all over the world.

Phylum	Arthropoda
Class	Arachnida
Sub-class	Chelicerata
Class	Arachnida
Order	Araneae

• Body is divided into two main divisions (Prosoma and Opisthosoma)

• Connected by pedicel

• Prosoma is dorsally covered by carapace

• Ventrally covered by two chitinous plates (small labium and large sternum)

• Prosoma bears eyes and six pair of appendages

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• Chelicerae are provided with poisonous gland

• Four walking legs having 7 segments

Phylum Arthropoda

Sub-Phylum Chelicerata

Class Arachnida

• Opisthosoma contains spinnerets, Female genital organ, book lungs

• May be nocturnal or diurnal

• Wolf spiders chase their prey

• Sit and wait predators

• Jumping spiders has largest eyes and keenest vision

• Crab spiders lurk among flower petals and fallen flowers, in wait of prey

• Mostly feed on insects

• Survive without food for many months

• Only few species are dangerous to man (i.e. Latrodectus)

• Power of regeneration (leg)

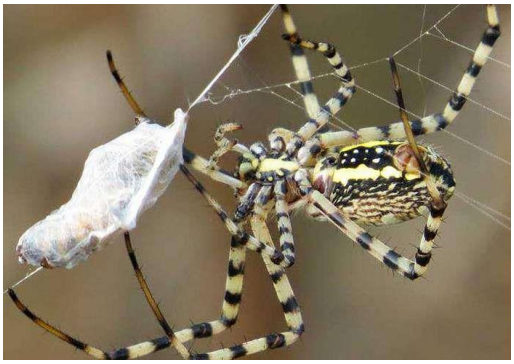
• Sexual dimorphism (male smaller)

• Female suck body sap of male after mating

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• Some species carry cocoon on the back of abdomen

Module# 161 Taxonomy and Ecology of Lady beetle



• Ladybugs, also called lady beetles or ladybird beetles, are a very beneficial group.

• They are natural enemies of many insects, especially aphids and other sap feeders.

• A single lady beetle may eat as many as 5,000 aphids in its lifetime.

• Adult lady beetles have very characteristic convex, hemispherical to oval shaped bodies that can be yellow, pink, orange, red, or black, and usually are marked with distinct spots. This is a type of warning coloration to discourage other animals that may try to eat them.

• Like many other brightly-colored insects, they are protected by an odorous, noxious fluid that seeps out of their joints when the insects are disturbed.

• Adult females usually lay clusters of eggs on plants close to aphid, scale, or mealybug colonies.



• The alligator-like larvae are also predators.

• They are spiny and black with bright spots.

• Although they look dangerous, lady beetle larvae are quite harmless to humans.



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• After feeding on insect prey for several weeks, the larva pupates on a leaf.

• Some lady beetle species have several generations each year while others have only one.

• During the summer months, all stages often can be found at the same time.

• Adults of some species spend the winter clustered together in large groups under leaf litter, rocks, or other debris.

• Important biological control agents

• Should be conserved

• Learn to recognize the different stages of these beneficial insects.

• Make insecticide applications only when necessary and use selective insecticides or limited treatments to avoid killing lady

• Birds are ladybugs' main predators, but they also fall victim to frogs, wasps, spiders, and dragonflies.

• Ladybugs are happy in many different habitats, including grassland, forests, cities, suburbs, and along rivers.

• Highly susceptible to pesticides

• Ladybugs are most active from spring until fall. When the weather turns cold, they look for a warm, secluded place to hibernate, such as in rotting logs, under rocks, or even inside houses. These hibernating colonies can contain thousands of ladybugs.

• 5000 species/different consumption rate

Module# 162 Taxonomy and Ecology of Green lacewing

• Green lacewings are an often under-appreciated group of beneficial insects.

• As with lady beetles, these natural enemies are important predators of many types of soft bodied insects and insect eggs.

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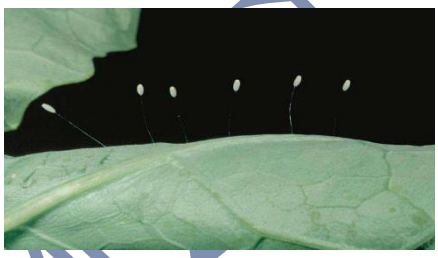
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• The adult green lacewing is about 3/4 inch long, light green and has a delicate appearance with lacy wings.

• One unusual characteristic are its eyes, they look like two golden hemispheres.

• They are weak fliers and are commonly found near aphid colonies.



• The adults feed mostly on nectar, pollen, and honeydew but with some species the adults will feed on insects.

• The eggs are either laid singly or in small groups. Each is always found perched on the tip of a hairlike stalk that is about **1/2 inch** long.

• This helps to reduce cannibalism of the eggs by sibling larvae.



• Females will usually deposit the egg close a food source for the larvae.

• The larvae are **brown and white** and may grow up to about l/2 inch in length.

• Larvae are called **aphid lions**, because they feed on other soft bodied insects as well as aphids.

• They are voracious feeders, attacking with large, curved, hollow mandibles.

Module# 163 Taxonomy and Ecology of Stink bug

• Pentatomidae is a family of insects belonging to the order **Hemiptera**, generally called shield bugs or stink bugs.

• Pentatomidae has over **4700** species

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• Have piercing sucking mouthparts

• Most are phytophagous, severe pests on agricultural crops.

• However, some species, particularly in the subfamily Asopinae, are predatory and may be considered beneficial.



• They have ability to release a **pungent defensive spray** when threatened, disturbed, or crushed.

• The composition of this spray may vary between species, and even by sex or age

• But generally includes **aldehyde and alkanes**.

• Moderate to large size, ranging in length from **7 to 25 mm**, and are broadly **elliptical** in shape.

• The subfamily Asopinae belongs to the family Pentatomidae of the suborder **Heteroptera (true bugs).**

• About **300 species** in **69 genera** have been described worldwide.

Taxonomy and Ecology of Green lacewing

• Adult stink bugs are strong fliers and typically migrate into cotton from alternate hosts outside the field, including adjacent fields of **maturing corn and soybean**.

• Stink bugs are difficult to scout.

• Adults tend to aggregate, and the distribution of stink bugs within a field may be highly concentrated, particularly along borders.

• Despite being relatively large insects, they can be difficult to see. Stink bugs can be scouted for visually, but most thresholds are based on drop cloth samples or the occurrence of internal symptoms of feeding damage to bolls.

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• Predatory stink bugs are a gardener's friend, feeding on more than **100 species** of insect pests. Both nymphs and adults attack insects larger than themselves, and suck the body fluids from their prey with a **needle-like beak**.

• Predatory stink bugs tend to have a **shorter, stouter beak** than the long, thin beak of plant feeding stink bugs.

• Adults overwinter in sheltered locations and leaf litter, emerging in early spring to stalk and devour their prey.

Taxonomy and Ecology of Stink bug

• **Life stage(s) that feed on pests**: Nymphs and adults.

• **Insect(s) fed on**: Spined soldier bugs prefer larger insects such as **grubs** and the larvae of various beetles and moths, and are especially fond of caterpillars. Two-spotted stink bugs often eat the **eggs and larvae** of Colorado potato beetle, and smaller caterpillars and larvae.

• **1st instar nymphs** of two-spotted stink bugs feed on **plant sap**, as will adults in early spring before prey are available, but do no significant damage to plants.

• **Where to find:** Generally throughout the garden, particularly on apples, asparagus, beans, cole crops, cucurbits, eggplant, potatoes, onions, sweet corn and tomatoes. .

• **How to attract and conserve: Pheromones** have been used to attract **spined soldier bugs** to the garden. Year round ground cover in garden will provide overwintering sites and shelter, as well as plant sap to feed on early in the season. Avoid or reduce use of broadspectrum **insecticides**.

Module# 164 Taxonomy and Ecology of parasitiod wasp

• Wasps, a very large evolutionary group of **hymenopteran insects,**

• Tamarixia radiata is a species specific **ectoparasitoid** of the **Asian citrus psyllid (ACP)**

Diaphorina citri

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• Diaphorina citri is an economically important **pest of citrus** in many citrus growing regions of the world (Halbert and Manjunath, 2004).

• The psyllid vectors the phloem limited gram-negative bacterium Candidatus Liberibacter Asiatic us, one of the causal organisms of the devastating citrus disease "huanglongbing" (HLB) or citrus greening disease. HLB is responsible for the decline of **most trees seen in disease** affected regions of the world

• **Tamarixia radiata** was first imported to **Florida from Taiwan** and **south Vietnam** and limited numbers were released soon after psyllid was discovered in the state (Hoy et al., 1999).

• Successfully managed the pest

• Saved citrus industry of Florida

• Tamarixia radiata was first described from **northwestern India (Punjab**) which is now part of Pakistan.

• Tamarixia radiata is not yet available commercially.

• The female usually deposits **one** and **very rarely two eggs** on the ventral side of the D. citri nymph, mostly in the groove between thorax and abdomen and close to the point of attachment of hind coxae.

• The **neonate larva** sucks fluid from the site of attachment

• The parasitized nymph remain indistinguishable from unparasitized nymph for **couple of days** until almost all body contents of the parasitized nymph are consumed and only wing buds and dorsal covering are left.

• At that point, the larva is mature and attaches itself to the leaf or shoot surface with a network of fine anchoring **silken threads**

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• Pupation takes place underneath the mummified nymph and the adult emerges by chewing a circular exit hole in the integument of the thoracic region.

• Development rate varies with temperature. It takes about **12 days to** go from egg to adult **at 25 ºC and 14L: 10D**.

• **A single female** can lay up to **300 eggs at 25-30 ºC** (Chu and Chien, 1991; Étienne et al. 2001).

PARASITIODS	PEST ATTACKED
***Ichneumonidae* wasps**	Larva of noctuids* (noctuids include helicoverpa, armyworms, cutworms and loopers)
Microplitis (WASP)	Helicoverpa larvae
Tachinid flies (Tachinidae)	Larva of noctuids
Trichogramma Wasp	Eggs of noctuids
Chalcid wasps	Aphids
Braconid wasps	beetles, caterpillars, flies and sawflies.

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Module# 165 Taxonomy and Ecology of Whitefly

• There **are1550 species worldwide** under two subfamilies.

• 70 in Punjab Pakistan

• **Cosmopolitan** insects

• Most abundant in the **tropical and subtropical** countries of the world.

Why called Whitefly?

• White appearance of adults due to **deposition of wax** on the on body and wings

• Adult whiteflies are **small, approximately 1-2 mm** in length.



• Adults with **wingspan less than 3mm.**

• With a **yellow** body and **two pairs** of white wings.

• The compound eyes **are red**.

• Sucking and piercing mouthparts.

• Both sexes have functional mouth parts and two pairs of functional membranous wings.

• Nymphs are initially semitransparent, later becoming creamy yellow to pale-green.

• The pupae or fourth nymphal stage bear **7 pairs of setae** on their dorsum. It is called **pseudopupa**.

• Produced large quantity of honeydew

• The honeydew emerges from the **anus**

TAXONOMY

Phylum	Arthropoda
Class	Insecta
Order	Hemiptra
Family	Aleyrodidae



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• Present under surface of the leaves.

• Both nymphs and adults feed.

• It has **160 host plants**.

• Polyphagous feed on a large number of host plants like Field crops, Fruit trees, Vegetables, Ornamentals , Weeds etc

• One of the most noxious insect pest attaching fields crops.

• Loss is due to activities of **sucking the sap from the leaves** that weaken the plants, resulting in low fruiting with reduced yield.

• They excrete large amounts of honeydew and throw it on different parts of plants.

• Honeydew **attract sooty mold** which affect photosynthesis

• They also act as vectors of many plant viruses, which adversely affect the growth and yield of crops.

Cotton leaf curl virus (CLCuV) disease in Pakistan. Sweet potato mild mottle virus. Yellow dwarf disease. They inject toxic saliva which results in plant disorders

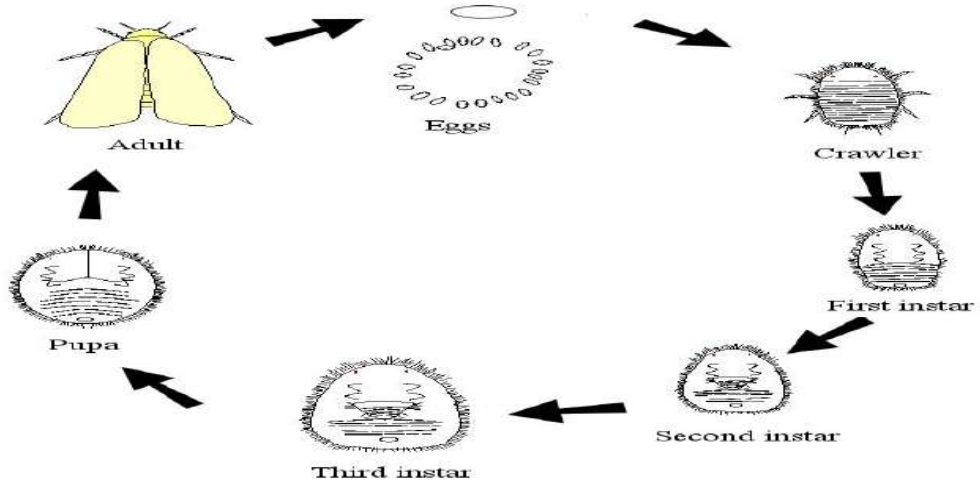
• Females can lay **10-150 eggs in her lifetime**, often in fairy rings • Eggs hatch into mobile crawler after **6-8 days**. • Crawlers settle at feeding site and lose the ability to walk.

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They insert their mouthparts into the leaf to feed on sap.



- **18-28 days** in warm months
- Adults can live **three weeks** and are often found in meeting pairs.
- Females are larger than males
- Resistance against pesticides.
- Rotation of insecticides

Biological control

- Small birds
- Spiders
- Lacewings
- Hoverflies
- Ground beetles
- Mirid bugs
- Damsel bugs and pirate bug

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Module# 166 Taxonomy and Ecology of Mealy bug

Phylum	Arthropoda
Class	Insecta
Order	Hemiptera
Family	Pseudococcidae
Genus	Phenacoccus
Species	*P. solenopsis*



• Found in all parts of worlds

• **Native to USA (Tinsley 1898)**

• Mostly found in **moist, warm climates**

• Many species are considered pests as they feed on plant juices

• Act as a vector for several plant diseases.

• Vehari in 2005

• Now almost in all cotton growing areas

• Affecting crop yields

• In 2006 (Multan, Bahawalpur, Kanewal, Vhari) as epidemic

• Mealy bugs are **sexually dimorphic**. Females are about **3 to 5 mm** long. Their body is usually covered with a **mealy waxy secretion**, often extended into lateral and terminal filaments. They usually remain clustered around the terminal shoots, leaves or fruits.

• They are **wingless** and do not move unless disturbed. They often retain **legs** and can move. Females are provided with **9 segmented antennae**.



• There are **18 pairs** of very short, lateral wax filaments, and several slightly longer caudal filaments

• Female Mealy bug **eggs are very small** and are laid under a white, loose woolly wax, which remains attached to the abdomen of the females. A female may lay between **50 to 600** eggs.

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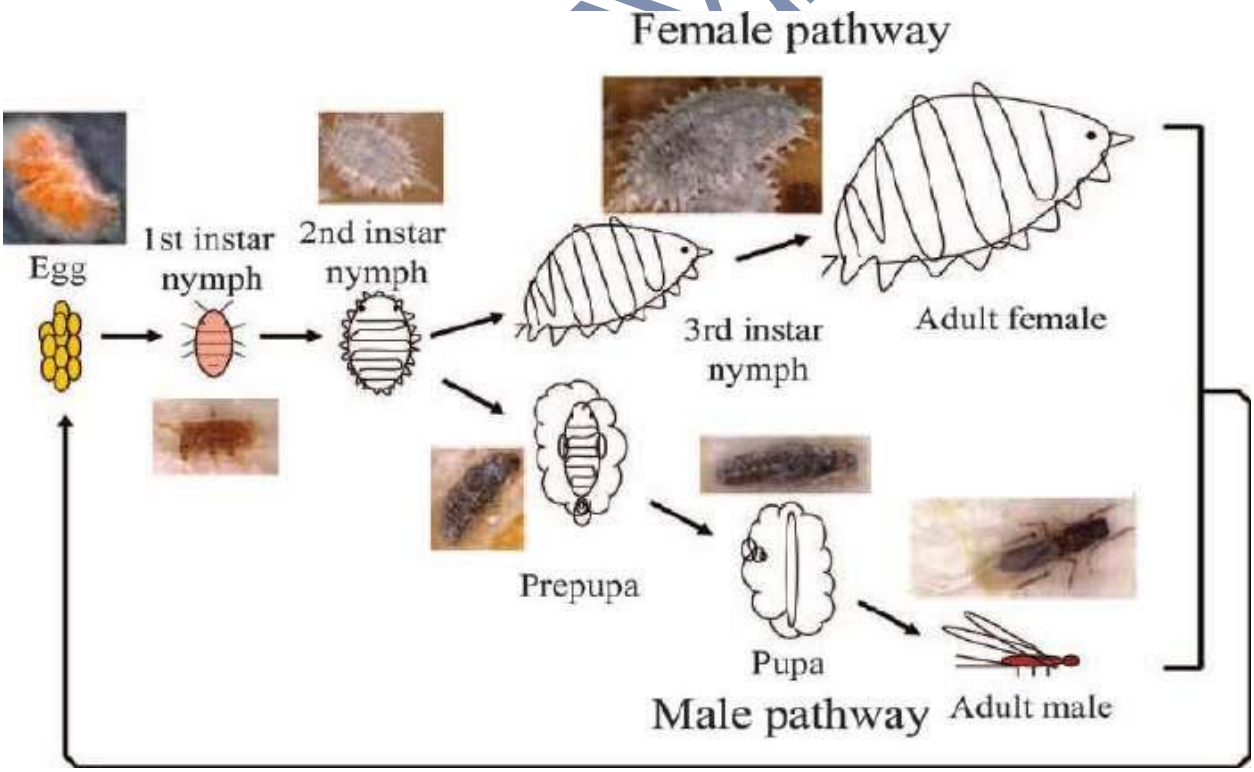
• Within **3-9 days** crawlers come out and actively move and spread out and settle down for on plant parts and gradually develops waxy coating.

• **Males are smaller**, gnat-like and **have wings**.

• Male mealy bugs do exhibit a radical change during their life cycle, changing from **wingless**, ovoid nymphs to wasp-like **flying adults.**

• Females feed on plant sap\Honey dew----Black sooty Moulds---**Photosynthesis:** • They attach themselves to the plant and secrete a powdery wax layer used for protection while they suck the plant juices.

• Honey dew---Ants—dispersal from plant to plant



• The **males** on the other hand are **short-lived** as they do not feed at all as adults.

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• **They only live to fertilize the females.**

• The most **venerable stage** of the cotton mealy bug is larval stage.

• It can be control easily by applying different control measures.

• The **control measures** of cotton mealy bug **are the traditional, biological, and chemical.**

Phylum	Arthropoda
Class	Insecta
Order	Lepidoptera
Family	Pyralidae
Genus	Scirpophaga
Species	*S. incertula*

Module# 167 Taxonomy and Ecology of Yellow stem borer



• **Serious pests of rice**

• Males are **ochreous (natural soil colour, clay**)

• The **spot** on the wings is **inconspicuous**

• Anal tuft is absent

• Female are pale yellow with distinct dark spot on the front wings.

• Tip of the abdomen is equipped with tuft of white and brown colored hairs.

• **Sexual dimorphism** in this insect has often led to the conclusion that 2 different species are involved.

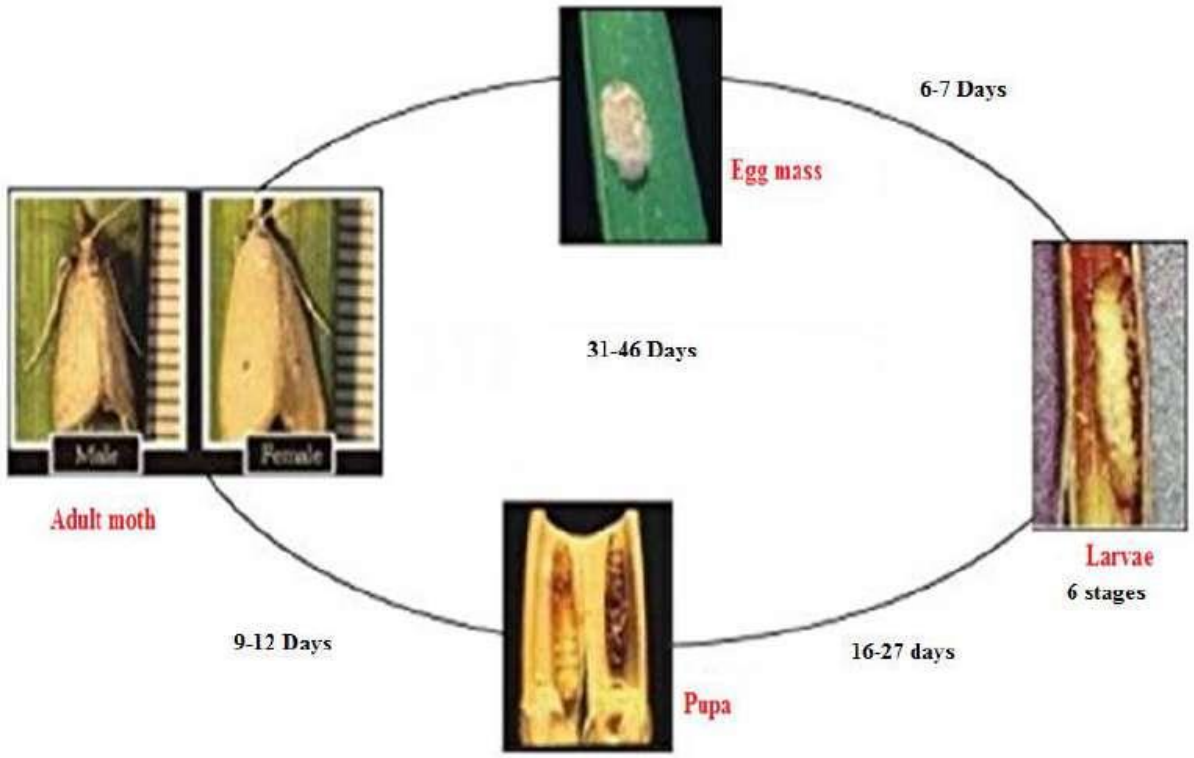
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• Adults are active after **dusk, mate** • Female lay eggs **120-150 underside** leaves • **2-5 clusters**





• The larvae first feeds on the inside of leaf-sheath and bore into the plant stem as a result of which the central shoot of the young plant dries up causing **"dead hearts".**

• In older plants the whole ear heads become **whitish and chaffy (white head**).

• Loss caused to rice production by this very pest in different regions and seasons varies from **5-90%.**

Control

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• Destruction of weeds harbouring pests

• Pulling out infested rice plant

• Flooding the field after harvest

• Stubble (short cut stalks) left after harvest should be ploughed up and burn or submerged in water immediately after harvest

• Crop rotation

• Application of insecticides

• Collection and destruction of egg masses from nursery plants.

• Setting up of light traps to trap and destroy the moth.

• Destruction of seedlings with "dead hearts" while planting.

Phylum	Arthropoda
Class	Insecta
Order	Lepidoptera
Family	Noctuoidea
Genus	Helicoverpa
Species	*H. armigera*



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Module# 168 Taxonomy and Ecology of American Bollworm

✓ ***American Bollworm (Very Important Question)***

IT is serious pest of cotton. • It is distributed in Pakistan • Central and southern Europe. Temperate Asia and Africa

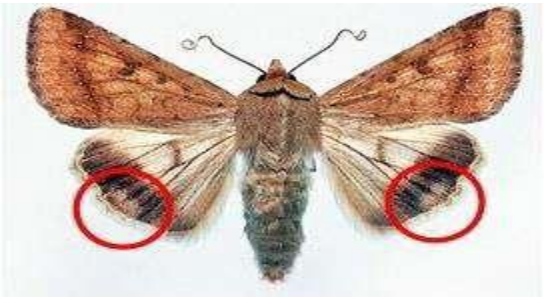
• The body length varies between **12 and 20 millimeters** with a wingspan of 30–40 millimeters.

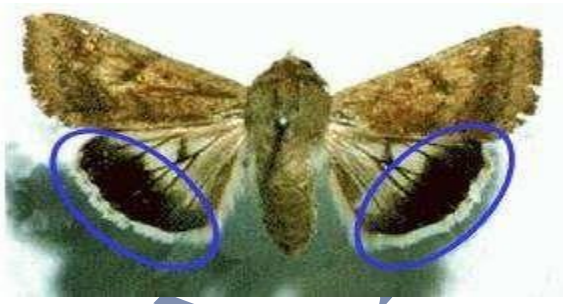
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• The fore wings are **yellowish to orange** in females and **greenish gray in males**

• The hind wings are a pale yellow with a narrow brown band at the external edge and a dark round spot in the middle.

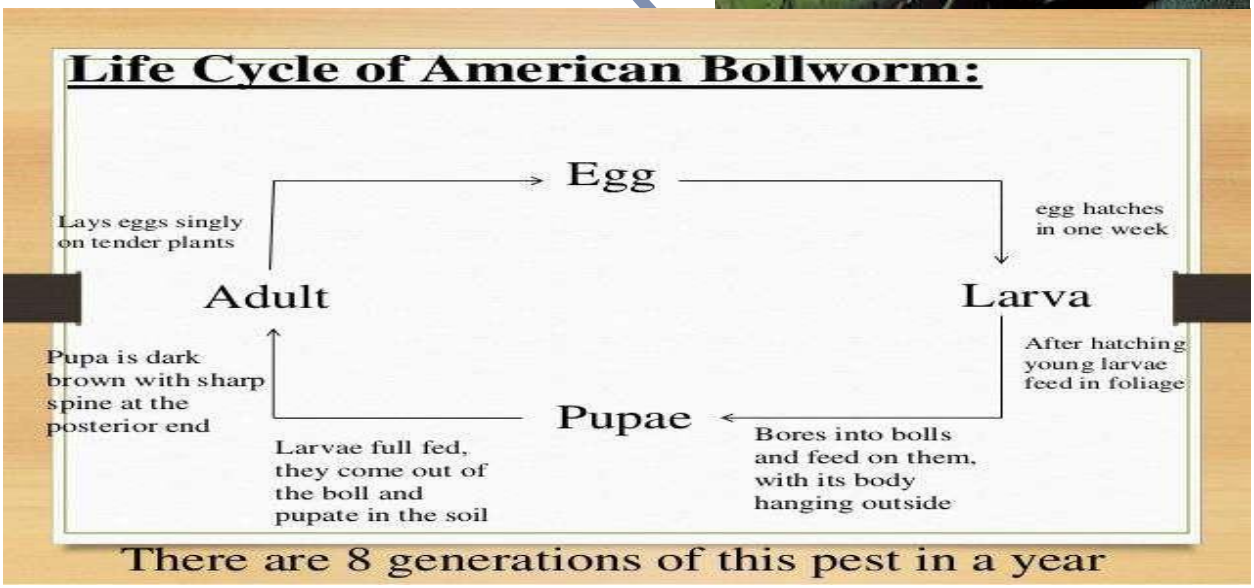




LIFE CYCLE OF AMERICAN BOLLWORM:

(Very Important Question)





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• Caterpillar are polyhagous. Feed on foliage and on the bolls of cotton. Reduce yield. Show prominent hole. Dirty feces accumulate on the boll surface. It shifts from one boll to another and **damages dozens** of bolls during life time.

• Avoid the cultivation of okra, moong and castor in and around the cotton crop

• Avoid growing of American cotton in the field

• Ichneumonid wasp, larval parasitoid of American bollworm should be promoted in the field. Spray profenofos EC 800 ml/acre

Module# 169 Principles of Systematics

Phylum	Arthropoda
Class	Insecta
Order	Lepidoptera
Family	Noctuoidea
Genus	Pectinophora
Species	*P. gossypiella*

Taxonomy and Ecology of Pink Bollworm

• The pink bollworms prefer areas where **cotton and okra** are being produced. • They originated in the **eastern Indian**

• All over the worlds

• Adult pink bollworms are **small and grayish-brown** in color with folded wings and a slender body.

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• Young larvae are small caterpillars with **dark brown heads.** • Once the larvae have matured, they are about **12 mm** in length with pink bands on their backs.

• Once laid, bollworm eggs take three to four days to hatch.



• Freshly hatched pink bollworm larvae are white with brown heads, turning pink in the fourth larval stage.

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• Larvae bore into cotton plants to feed on their seeds. • The larvae take **twelve to fifteen days** to develop, then they move to the soil to pupate. • The brown pupa is immobile for **its first seven to eight days.**

• Common signs of an infestation include shedding of flower buds, deformation of flowers, reduction of lint development and destruction of the seeds in lint.

Phylum	Arthropoda
Class	Insecta
Order	Hemiptera
Family	Cicadellidae



• Bollworms' greatest harm is in the damage they do to cotton bolls. • As the larva of the pink bollworm burrows into the boll, its stains the lint, rendering it commercially unusable. • If the humidity is high, it only takes **one or two larvae** to completely destroy a boll.

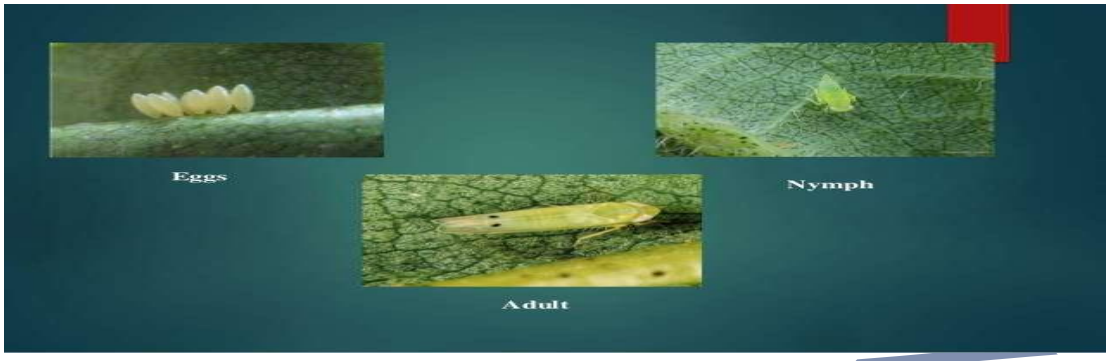
Module# 170 Taxonomy and Ecology of Jassids

• Major sucking pests of cotton.

• Eggs yellowish white.

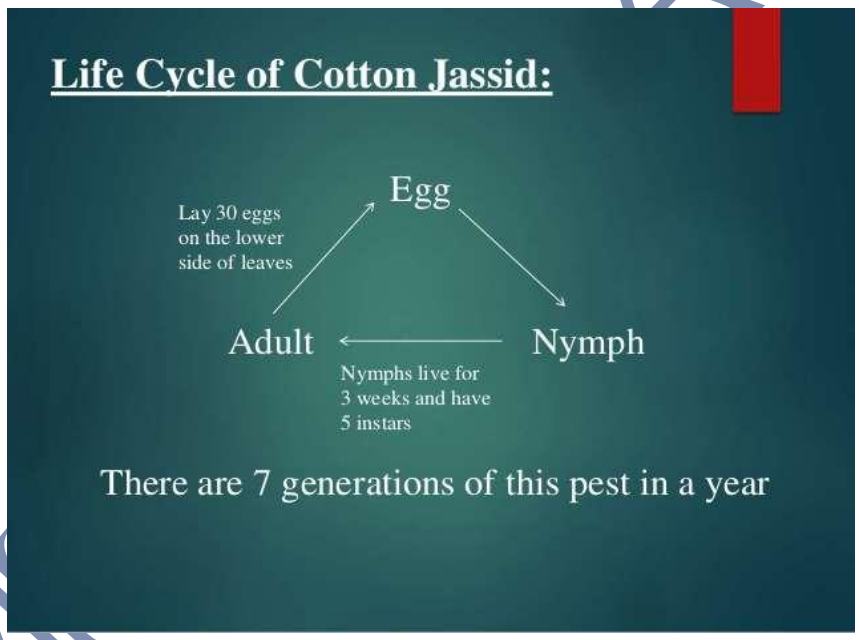
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LIFE CYCLE OF Jassids:

(Very Important Question)



- Nymphs and adults are greenish yellow with a black spot on tip of each forewing. Jump on disturbing. Nymphs and adults both suck sap.

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Leaves first show yellow and then reddish spots.



• Attacked leaves turn downwards, dry and fall • Plants become weak and their fruit fall

Control

• Clean cultivation and destruction of alternate host plant

• Selected varieties of cotton should be grown

• Green lace wing is a good predator of cotton jassid

• Spray of **acetamiprid 125ml/acre**

• Spiders

Module# 171 Taxonomy and Ecology of Thrips

Phylum	Arthropoda
Class	Insecta
Order	Thysanoptera
Family	Thripidae

• Tiny sucking insects

•Their identification to species by standard morphological characters is often challenging.

•There is also considerable variability leading to many species being misidentified. Molecular sequence based approaches have increasingly been applied to their identification

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• **Fringed wings** • Also transmit **tomato spotted wilt virus** to many plants

• Feed on wide range of plants world wide

• They prefer tender parts of the plant, such as **buds**, **flowers** and new **leaves.**

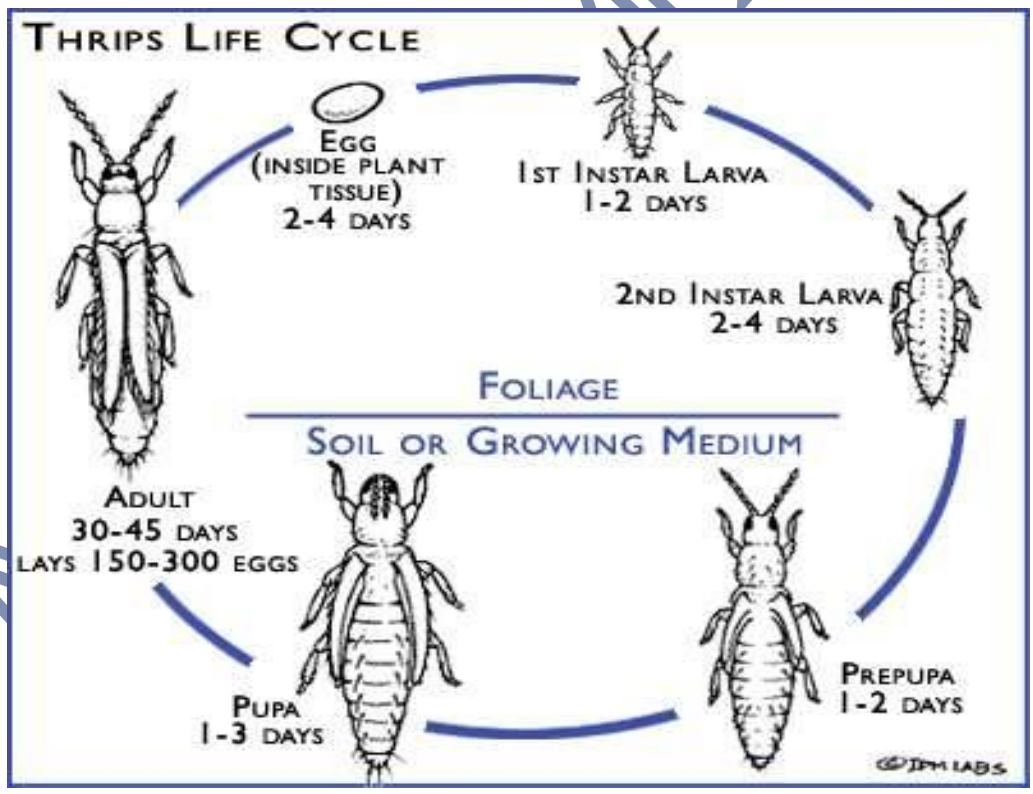
•Besides feeding on plant tissues, the common blossom thrips feeds on pollen grains and on the eggs of mites.

•When the larva supplements its diet in this way, its development time and mortality is **reduced**.

Adult females that consume mite eggs increase their fecundity and longevity.

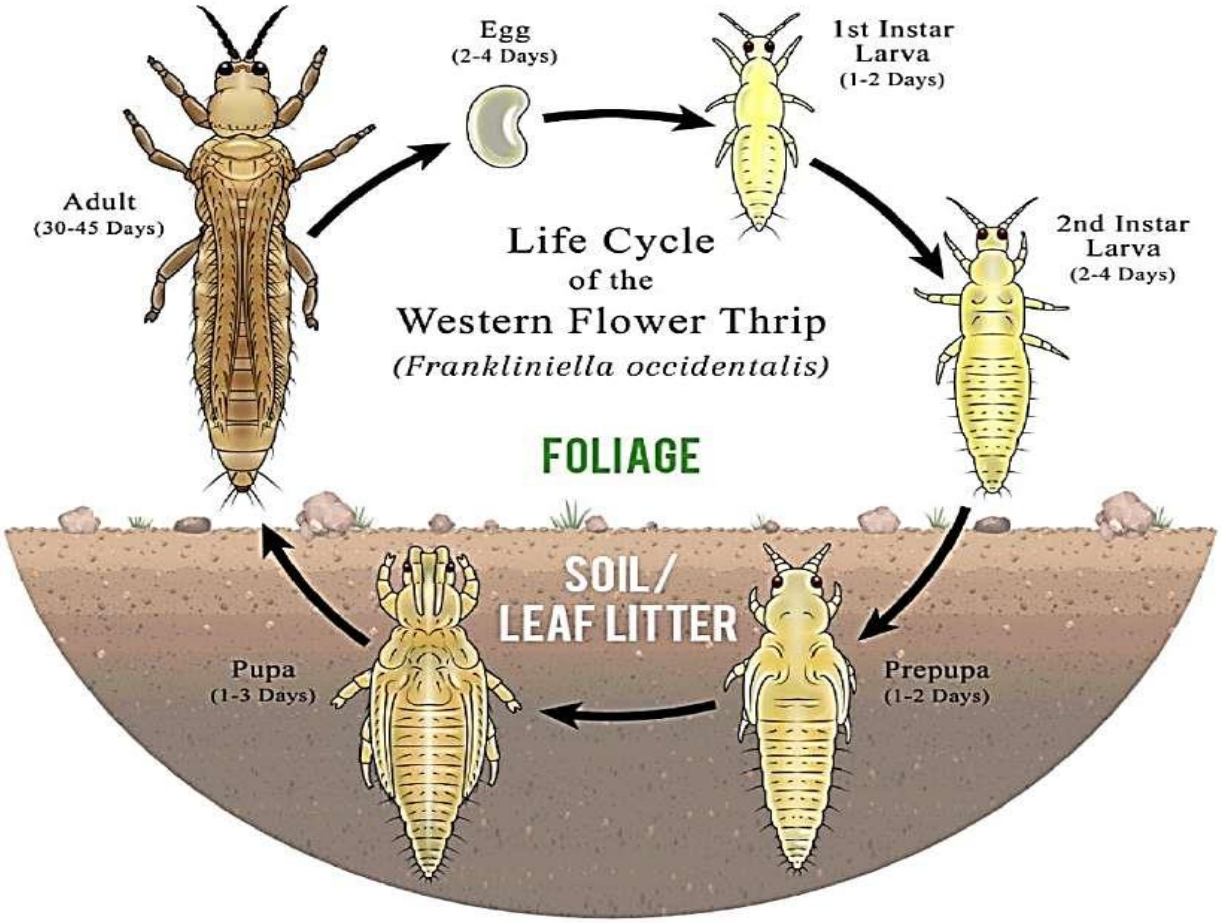
LIFE CYCLE OF Thrips:

(Very Important Question)



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Damage

•In the right conditions, such as in greenhouses, many species can exponentially increase in popu lation size and form large swarms because of a lack of natural predators coupled with their abilit y to **reproduce asexually**

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• Sucks plant cells from many garden flowers, plants and fruits

• Damage includes **streaks** and **small white patches**

• Major infestation leads to stunted growth and damaged fruit and flowers

Control

• Try using **yellow** or **blue sticky traps**

• Spray **dormant oil** on fruit trees

• Shake branches of plants to remove the thrips

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• Spray of insecticidal soap kills them

• Reflective mulches can be used



Module# 172 Taxonomy and Ecology of Rice Leaf Folder

• Very active

• Major pest of rice

• Bright yellow or straw in color

• Two wavy lines in forewing and one in hind wing

• Eggs are laid singly or in groups in rows beneath the leaf surface

• Eggs are scaly white in color

• Incubation period is 4-8 days

Phylum	Arthropoda
Class	Insecta
Order	Lepidoptera
Family	Crambidae

Damage

• Larval stage is damaging

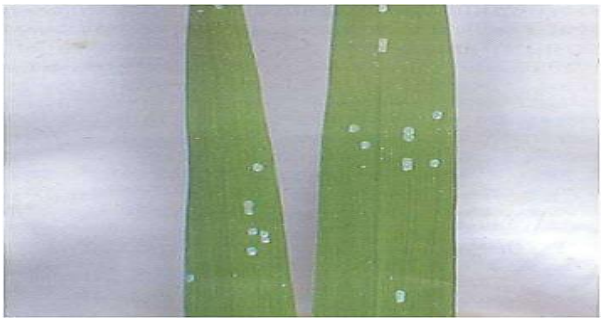
• All stages of plants are damaged by this pest

• Newly hatched caterpillar cut leaf edges and folds the leaf

• Infested leaves appear white

• Attacked plant dries up

• Yield loss vary from 10-50%



Control

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• Reform tillage and cultivation system

• Light trap or lure can be used.

• Utilize natural enemies (beetles and Trichogramma spp.) as biological control

• Application of chlorpyrifos EC 1000ml/acre

Module# 173 Taxonomy and Ecology of Citrus Leaf Miner

• Origin-Asia

• Distribution-worldwide

• Leaf miner: when the larvae being very small, live in between the two epidermal layers of leaves and feed on the food material inside.

• Examples: Citrus leaf miner, the cashew and mango leaf miner, and the buprestid leaf miner etc.

Phylum	Arthropoda
Class	Insecta
Order	Lepidoptera
Family	Gracillariidae

• Eggs whitish in color usually laid towards the petiole of leaf

• Larvae are translucent, light green in color and very difficult to detect.

• The pupa is yellowish-brown and darkens with age

• Adults females have a black and yellow color with a conspicuous yellow dorsal spot.



Damage

• Leaf mining results in



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• Leaf deformation

• Leaf chlorosis

• Necrosis

• Leaf fall

Control

• **Monitoring:**

Pheromone traps can be used

• **Biological control:**

General predators and parasitoids

• **Chemicals**:

q Abamectin

q Imidacloprid

q Spinosad

Module# 174 Principles of Systematics

Taxonomy and Ecology of Citrus Psylla (Diaphorina citri)

Phylum	Arthropoda
Class	Insecta
Order	Hemiptera
Family	Liviidae

• Newly laid eggs are pale brown but the turn yellow and finally orange before hatching

• Nymph is generally yellowish- orange

• Adults are mottled brown

• Body is covered with whitish waxy secretion, making it appear dusty

• Greening disease or Huanglongbing

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Damage

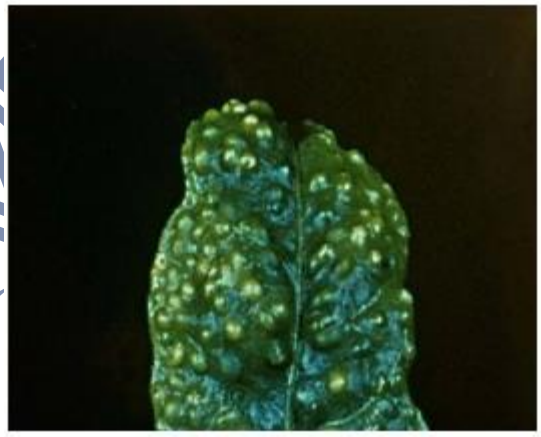


• Vectors of causative agent of Citrus greening disease or Huanglongbing

• Psyllid nymphs produce a toxin that causes the plant tips to die back or become contorted, preventing the leaves from expanding normally.

• **Control**:

Traditional control:



Infected plants and trees should be eradicated before spread of disease

• Chemical control:

Insecticides Injection of trees with tetracyclin and antibiotics

• Biological control:

§ Maggot Fly Larvae

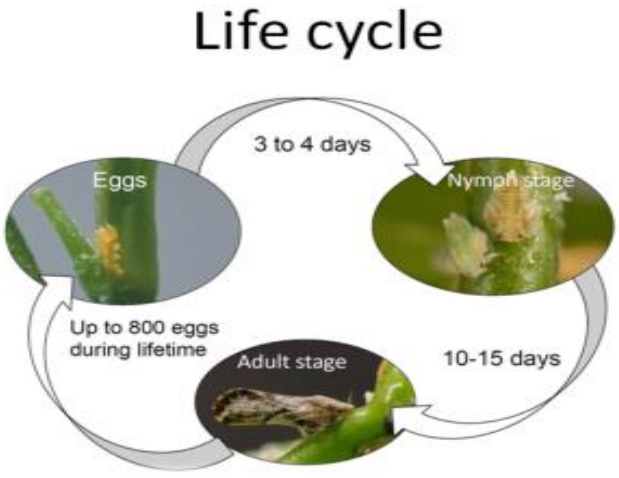
§ Lady beetle

§ Tamarixia radiata

§ Lace wing

§ Pterrastichus radiates

LifeCYCLE:



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Module# 175 Taxonomy and Ecology of Aphids

Phylum	Arthropoda
Class	Insecta
Order	Hemiptera
Superfamily	Aphidoidea

• Soft bodied-weak

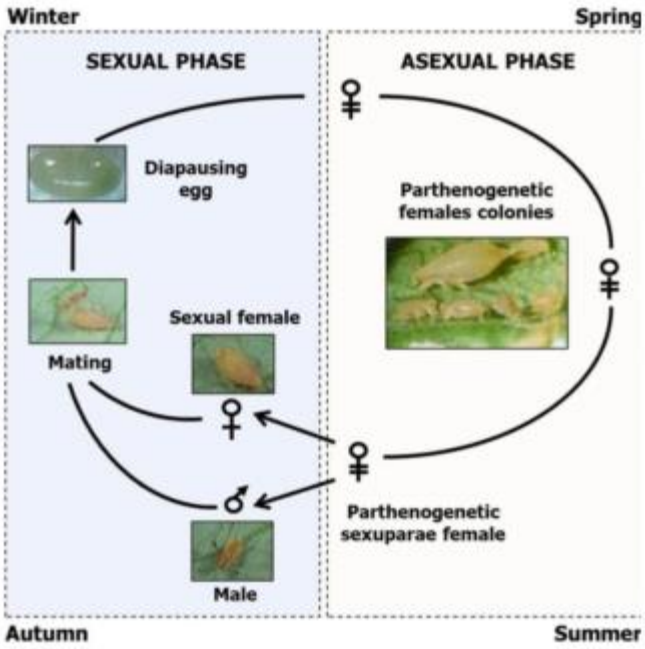
• Compound and simple eyes

• Walking legs:skinny

• Head pointed downwards for feeding

• Most aphids have conicles (excrete pheromones)

• Also have cauda (flick honeydew)



Feeding habits

• Aphid feed from the phloem of plants which they trap into with the styletes of their probosocis.

They gain access to the phloem vessels from 3 main parts of the plant, stems, leaves, and roots.



Damage

• Large populations of aphids can turn leaves yellow and stunt shoots; aphids can also produce large quantities of a sticky exudates known as honeydew, which often turns black with the growth of a sooty mold fungus

Control

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Chemical control

• Using insecticides like carbamates, pyrethroids and organophosphates

Biological control

• Ladybug, Ladybug larvae,

Lacewing, Parasitoids wasp,

Mummies, Fungal Pathogen

Module# 176 Phenon, Taxon and Category

Phenon

A term used for the different forms or phenotypes that may occur within a single population.

It includes:

• Sexes

• Age stages

• Seasonal varieties

• Morphs

Phenon

• The term Penon was first introduced by the Camp and Gilly (1943) to describe phenotypic ally homogeneous samples at the species level.

• It was later used by Sneath and Sokal (1973)

Taxon

• Defined by simpson (1961) as 'a group of real organisms recognized as a formal unit at any level of hierarchical classification'

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• In other words

• A taxon is any rank

Taxon

• In other words A taxon is any rank that is considered sufficiently distinct by the taxonomists to be formally recognized and assigned to adefinite category.

Two aspects:

1. A species is not a taxon but category• Example: Robin (Tundus migratorius) is a taxon
2. Taxon: Taxon must be formally recognized by the taxonomists Examples: Demes Geographic isolates within a species become taxa only –subspecies. Wiley (1981) called Taxa as historical groups which are clearly individuals like and not classes.

1. **Higher taxa:** Thrushes, Birds, Vertebrates

2. **Lower taxa:** Bluebirds, Rbins

Category: 'Rank or level in a hierarchical classification' a class whose members are all the taxa

• A class whose members are all the taxa that are assigned a given rank Examples:

The **species category** is a class whose members are species taxa.
20 or more categories fall into 3 categories

1. The species category

2. Infraspecific categories

3. Higher categories

Phyllum	Arthropoda
Class	Insecta
Order	Lepidoptera
Family	Bombycidae
Genus	Bombyx
Species	mori

Module# 177 Silkworm

• Silkworm, common name for the silk-producing larvae of any of several species of moths.

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• Silkworms possess a pair of specially modified salivary glands called silk glands, or sericteries, which are used in the production of cocoons.

Silkworm

• The silk glands secrete a clear, viscous fluid that is forced through openings, called spinnerets, on the mouthparts of the larva; the fluid hardens as it comes into contact with air.

• The diameter of the spinneret determines the thickness of the silk thread produced.

• The best-known silkworm is the larvae of the common, domesticated silkworm moth.

This moth, native to China, was introduced into Europe and western Asia in the 6th century AD and into North America in the 18th century.

• Silk is know as queen of textile due to sheen and luster

• Silk is one of the most beautiful and precious fibers given to us by nature

• It is considered bio-steel because of its strength

• China is the largest producer of silk followed by India

SERICULTURE IN PAKISTAN

• Despite the growing demand for natural silk, this industry could not flourish in Pakistan.

• This industry is facing numerous challenges such as lack of trained personnel, high mortality rate in local silk breed, and poor economic conditions of the farmers.

• Unfortunately, this industry has not been on the list of priorities of our Government.

• Provide livelihood for rural poor families but silk proteins also have wide range of applications in textile, food, cosmetics and pharmaceutical industry as well as in traditional medicines.

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• Currently, Pakistan is importing about 70% raw silk to meet the demand of local market.

Feeding

• Mulberry silkworm feeds on mulberry plants of genus Morus

• M. alba

• M. indica

• M. nigra

• Deep rooted plants

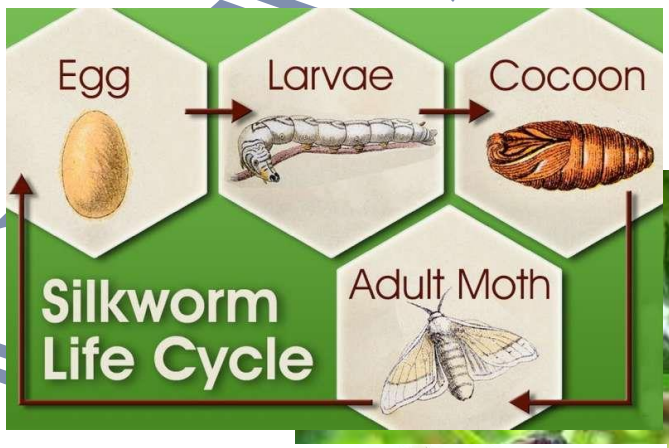
• Grow well in loamy soil or red sandy loamy

• Plantation during early spring and late autumn

• Mulberry plants grow 15-37 degree temperature

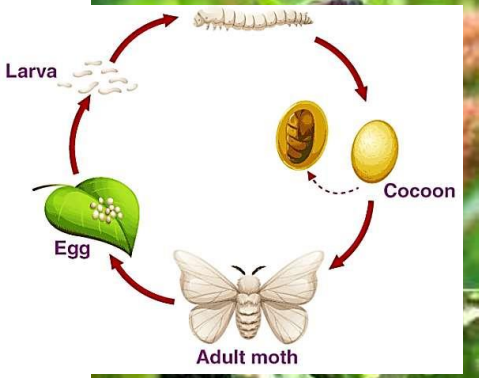
• 600-2500mm rainfall

• 300-900m altitude above sea level



Module# 178 Life Cycle of Silkworm

Copulation



• Following emergence from the cocoon, the females rest on the cocoon, spreading out her feeble wings and immediately begin to scent by exuding pheromone from her abdominal projection.

• The males being attracted by the pheromone find the female and start mating.



Eggs lying

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• Thus they remain over for 12-24 hours when the females get insemi-nated by the male.

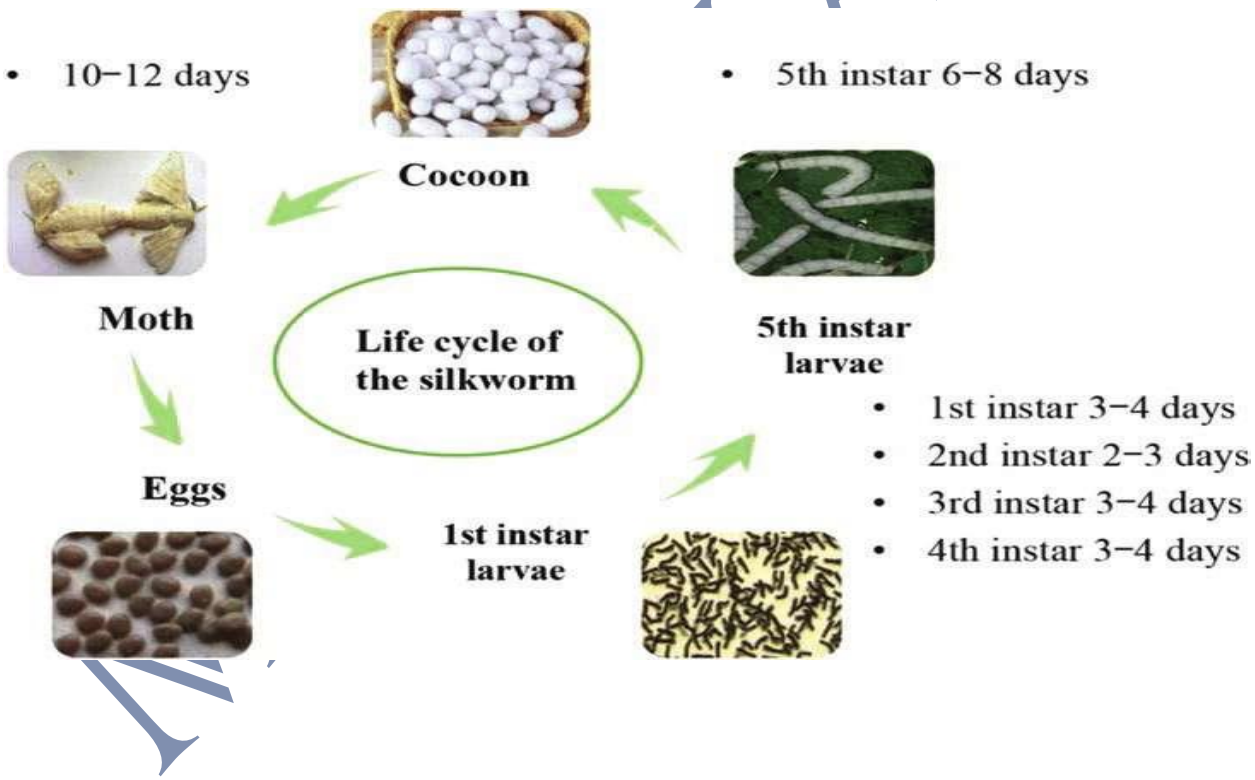
• After mating for 12-24 hours, the pair gets separated and the female starts laying eggs while the male moths die.

• Most females lay 150-300 eggs over the course of 1-2 days, some can lay as many as 1000 also.

• Fifth instar larvae makes quick round movements of its head

• Rate of 65 movements per minutes while spinning its cocoon.

LIFE CYCLE OF SILK WORM:



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Stages	Period (days)	Characters
Egg	9-12	Yellowish-white, semi round, 1 mg wt
1 instar	3-4	Last instar larvae is greenish, cylindrical caterpillar, 3-3.5 inches long, bear 3 pairs of thoracic legs, 5 pairs of pre-legs, last pair of leg modified to claspers, 1 thoracic and 8 abdominal spiracles, each larvae feed on 90 gm of leaves during larval period (voracious feeder)
II instar	2-3	
III instar	3-4	
IV instar	5-6	
V instar	7-8	
Prepupa	4-7	Non-feeding stage, cocoon spinning within 2-3 days
Pupa	10-12	
Adult	2-3	Robust creamy white, 2-3 day, bipectinate antennae, non-feeder, poor fliers, each female lays 400-600 eggs, secrete alkaline secretions to dissolve cocoon fibers.

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Module# 179 Composition of Silk

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Silk Composition

- Silk consists of two main proteins, sericin and fibroin, fibroin being the structural center of the silk, and serecin being the sticky material surrounding it.

Fibroin	70-75%
Sericine (silk gum)	25-30%
Waxy substances	2-3%
Natural colours	1-1.5%
Mineral matter	0.5-1%

Fig- Silk Composition



Fig -Raw Silk Microscope View

Application of SILK SERICIN:

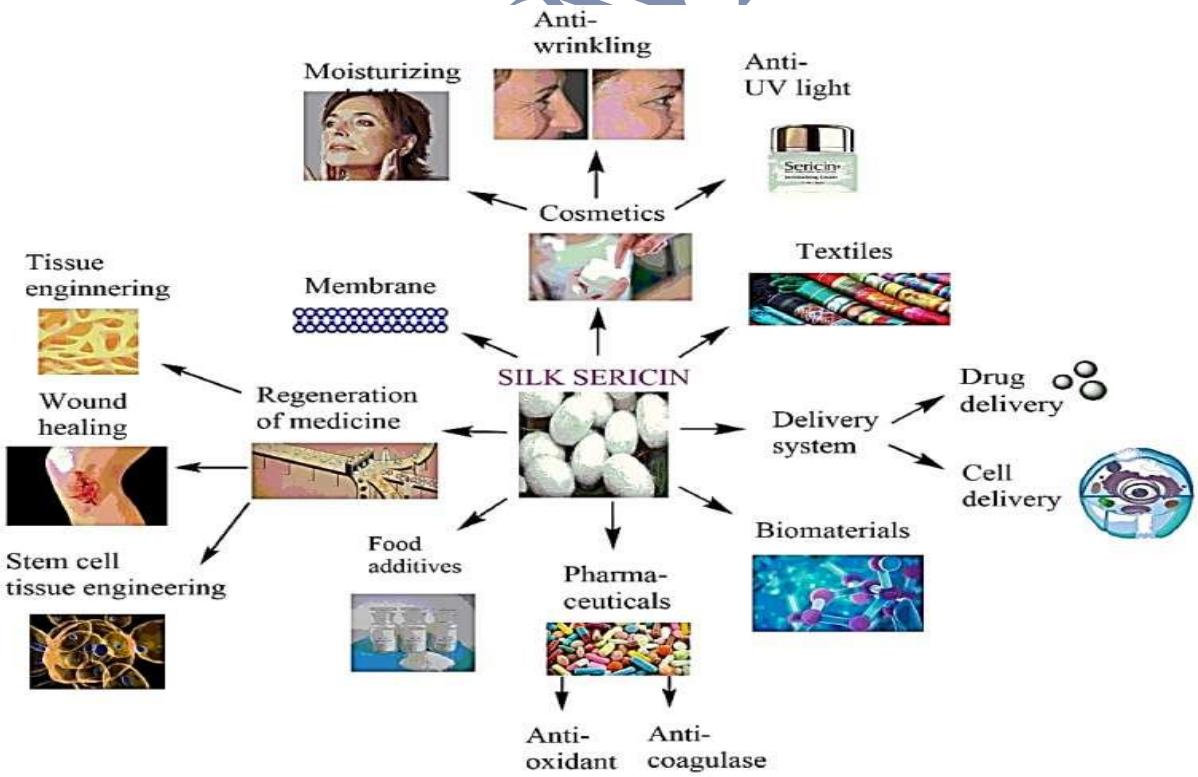


Figure 2. Applications of silk sericin.

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Application of SILK SERICIN:

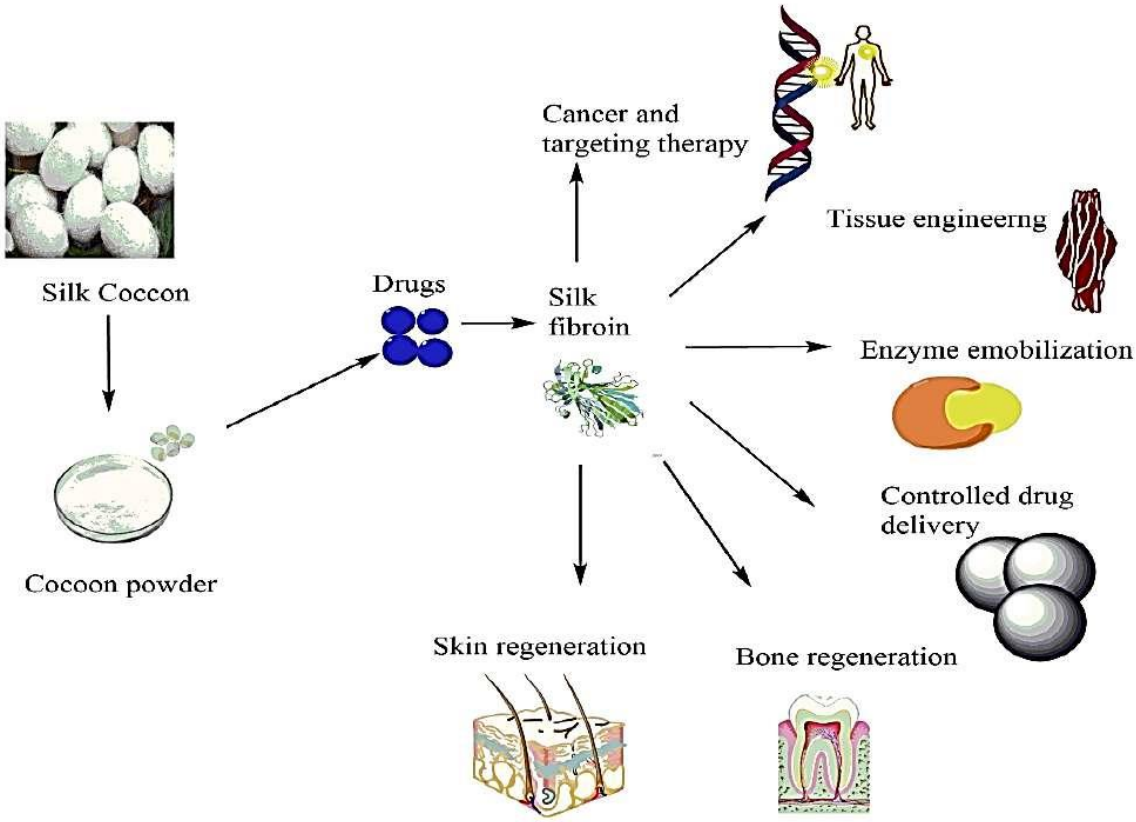


Figure 3. Applications of silk fibroin.

Module# 180 Lac Insect

• Lac insect live as parasite on sap of certain tress and shrubs and shows sexual dimorphism

• Babul (Acacia arabica)

• Khair (Acacia catcchu)

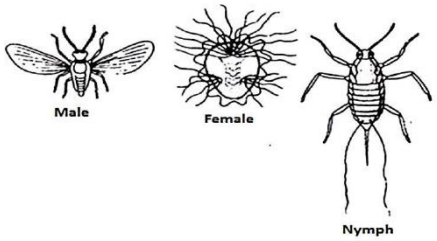
• Its life is completed in 6 months

• Two crops can be obtained

Kingdom	Animalia
Phylum	Arthropoda
Class	Insecta
Order	Hemiptera
Family	Kerriidae
Genus	Kerria
Species	*Kerria lacca*

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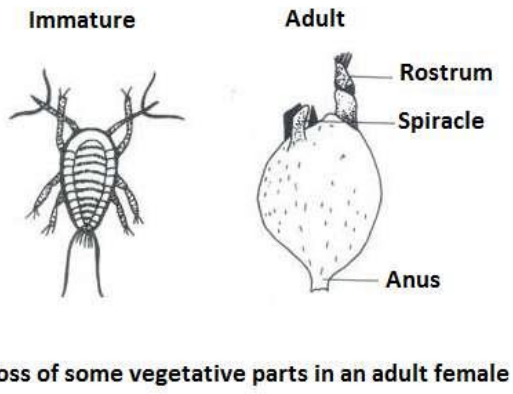
Sexual dimorphism

• The adult lac insect shows a marked phenomenon of sexual dimorphism. The male and female insect varies in:

• Shape

• Size and

• Presence or absence of certain body parts



Loss of some vegetative parts in an adult female

Female Lac Insect

• Small in size

• Head bear a single Antennae

• Single Proboscis

• Eyes are absent

• Female produce bulk of Lac

Fertilization

• After attaining the maturity, males emerge out from their cells and walk over the lac incrustations.

• The male enters the female cell through anal tubular opening and inside female cell it fertilizes the female.

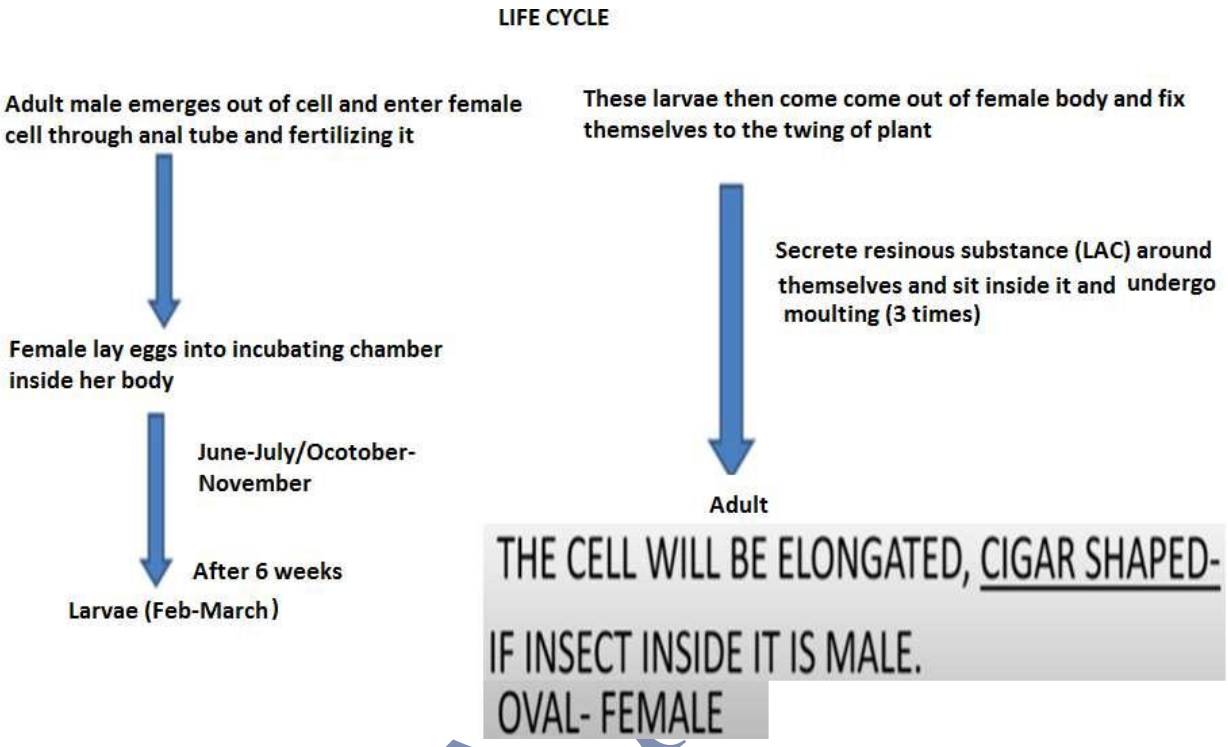
• After copulation, the male dies. One male is capable of fertilizing several females.

• Females develop very rapidly after fertilization.

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• They take more sap from plants and exude more resin and wax.



Module# 181 Lac Insect

Swarming

• These larvae begin to wander in search of suitable centre to fix them.

• This mass movement of larvae from female cell to the new off-shoots of host plant, is termed as "swarming".

Lac Secretion

1. Glands present in the abdomen of the lac insects.

2. Secrete immediately after the larval settlement on the new and tender shoots

3. This secretion appears first as a shining layer which soon gets hardened after coming in contact with air.

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4. This makes a coating around the insect and the twig on which it is residing.

5. As the secretion continues the coating around one insect meet and fuses completely with the coating of another insect.

6. In this way a continuous or semi-continuous incrustation of lac is formed on the tender shoots.

1. Proper care of host plant

2. Regular pruning of host plant

3. Propagation of insect

(Inoculation or infection)

4. Control of insect pests

5. Collection and processing of lac

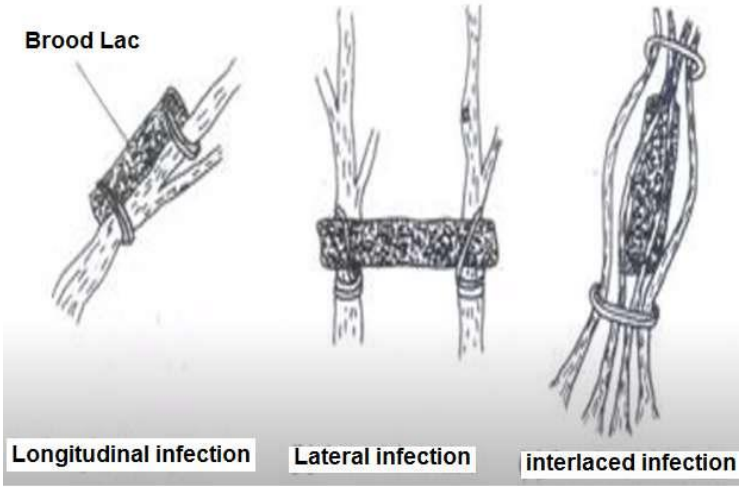
Inoculation

• The method by which the lac insects are introduced to the new lac host plant is known as inoculation.

• This may be of two types, namely “Natural infection” and “Artificial infection”.

• When infection from one plant to other occurs by natural movements of insect, it is called natural infection.

• This may be due to overcrowding of insect population and non-availability of tender shoots on a particular tree.



Module# 182 Lac Insect

Artificial Infection

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• Prior to about two weeks of hatching, lac bearing sticks are cut to the size of six inches. They are called **"Brood lac".**

• Brood lacs are then kept for about two weeks in some cool place.

• When the larvae start emerging from this brood lac, they are supposed to be ready for inoculation

Extraction of Lac

• Lac cut from the host plant is called as "stick lac".

• Lac can be scraped from the twigs before or after the emergence of larvae.

• Ari lac (if used before emergence of larvae)

• Phunki lac (if used after swarming of larvae)

• Swarming

Scraping of Lac

• The scraping of lac from twig is done by knife, after which they should not be exposed to sun. The scraped lac is grinded in hard stone mills.

• The unnecessary materials are sorted out.

• In order to remove the finer particles of dirt and colour, this lac is washed repeatedly with cold water.

• Now at this stage it is called as "Seed lac" and is exposed to sun for drying.

• Bleach lac is a refined product obtained by chemical process.

• Shell lac is dissolved in Sodium carbonate solution and bleaching the solution with Sodium hypochlorite. Precipitation of the resin with sulphuric acid.

• Deteriorate quickly and should be used within 2-3 months.

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Module# 183 Principles of Systematic

In 19 the century, term subspecies replaced the term variety

Ant specialists use term not only for geographic races but also for sibling species and individual variants.

'An aggregate of phenotypic ally similar populations of a species inhabitation a geographic subdivision of the orange of that species and differing taxonomically from other populations of that species.'

Temporal sub species identified on the basis of paleontology

Sub species

Difficulties in application of sub species

Wilson and brown (1953) and Inger (1961) pointed out four aspects:

1. The tendency of different characters to show independence trends

2. The independence occurrence of similar or phenotypic ally indistinguishable populations

3. The occurrence of micrographic races within subspecies

4. The arbitrariness of the degree of distinction-local population

'Reproductively isolated species without morphological difference are called sibling species'

• Sibling species are very similar in appearance, behaviour and other characteristics but they are reproductively isolated.

• In other words, sibling species are pairs or groups of genetically closely related species which are often morphologically indistinguishable, but are reproductively isolated, meaning that while they may interbreed, the offspring cannot reproduce.

Module# 184 Principles of Systematics

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Species that contain two or more subspecies are called polytypic species. Species that are not subdivided into subspecies are called monotypic species.

The term formenkries and rassenkries were formally applied to polytypic species

Polytypic species

• Discovery of the fact that previously morphospecies were not reproductively isolate and did not qualify as a separate species hence…sub species

• Polytypic species provide simplicity of classification of birds, snail, mammals and butterflies

EXAMPLES: 1900 monotypic species of birds----------9040 polytypic species

Importance of recognition of Polytypic species

1. Differentiation between allopatric and allochronic populations at subspecies level

2. Closely related species replace each other geographically

3. Species are on the boderline between sub species and species
(Species under the process of evolution (Incipient species)

Difficulties of Polytypic species

Occurrence of Polytypic species in animal kingdom

• In animals 40-80% are polytypic

• Formenkries and rassenkries are the order terms for polytypic species

Module# 185 Phenetics Vs Cladistics

• In the field of taxonomy there are two methods of constructing phylogenetic threes i.e.

Phenetics and Cladistics

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• Phnetics methods construct threes (phenogram) by considering the current states of characters without considering the evolutionary history that brought the species to their current phenotype.

• However, cladistics method depend on the assumptions about ancestral relationships as well as current state.

• Which character states are deeply conserved characters

• Which character states are not deeply conserved characters

• Conserved character state are used for phylogenetics reconstruction

• In cladistics approach trees are drawn based on conserved characters

• Phenetic approach trees are drawn based on some measurements of morphological traits.

• Molecular phylogenies are inferred from molecular data (DNA sequence, Protein sequence)

• Phenetics or cladistics or mixture of both

• Traditional taxonomy is based on phenetics.

• Phenetics is based on overall similarity to group organisms into hierarchical levels called Taxonomic groups.

• Faster and convenient but not accurate.

• Cladistics use certain shared features to group the organisms by common ancestor (better method of determining ancestry) • Linnaeus was a Phenetists

• Although he did not believe in evolution by descent but his pattern of classification provide a framework for thinking about evolution from common ancestor.

• For him "relatedness" meant propinquity in creators' design.

• Phenetic clustering

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• Darwin was a Cladist

• According to him natural system based on descent with modification.

• The characters that show true affinity have been inherited from a common patent.

• All true classification is genealogical

• Phenetic classification is artificial

• Cladistic classification is natural (Darwin, origin of species, 1859)

• Computer algorithms based on the phenetic model rely on distance method to build tree from sequence data.

• Phenetic approach lead to faster algorithms and have nice statistical properties for molecular data.

• Error rate is less

• But final tree is not highly reliable

Claisdtic

• Evolutionary relationships are documented by creating a branching structure (phylogenetic tree)

• It illustrates the relationships among sequences

• Cladistic method construct a tree (cladogram) by considering various possible pathways of evolution and chose from among the best.

• Evolutionary relationships are documented by creating a branching structure (Cladogram)

• Phylogram is a tree with branching that are proportional to the evolutionary distances.

• It is more informative

Module# 186 Homology

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What are homologous structure?

• A body part with a same structure between some groups due to inheritance from same ancestor.

• Not necessarily performing same function

Importance of homologous structure

• To provide a path of evolutionary paths

• Diverges between species and sub-species

Common examples

• Common example of homologous structures is the forelimbs of vertebrates, where the wings of bats and birds, the arms of primates, the front flippers of whales and the forelegs of four-legged vertebrates like dogs and crocodiles are all derived from the same ancestral tetrapod structure.

Tail bone

• The tail bone in human is a homologous structure to the beginning of many animals' tails, such as monkeys. It is known as a "vestigial structure" because it is the last vestige of what was once a tail. This structure serves as evidence of having a common ancestor, one that would have had a tail.

Neck bone

• In spite of its height, the giraffe has the very same number of neck bones (seven) as a giant whale and a tiny human being.

Pelvis bones

• Human beings, dogs, and cats all have similar pelvises, which are homologous structures to a vestigial pair of bones in snakes. These bones in snakes are the last remains of a pelvis, with no legs to attach.

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Eyes structure

• Human eyes are homologous to the eye bulbs which blind creatures who live in caves – like cave fish-- have on their heads.

Plasma membrane

• All organisms contain homologous plasma membranes with what is called a phospholipid bilayer.

Auditory bones

• An auditory bone that exists inside the ears of mammals is a homologous structure to a reptile's jawbone, including dinosaurs.

Stinger of wasps and bees and ovipositor of insects

• Wasps and bees have stingers that they can use when they feel they are in danger. This is a homologous structure to the ovipositor of other organisms like insects and certain fish; the feature allows these organisms to lay eggs.

Module# 187 ANALOGY

• The structure which are different in different species but perform similar function.

• Have similar appearance

• Not evolved together

• Do not share common ancestor

• Linnaeus started classification on the basis of morphological similarities.

• Wrong classification

• All similar looking organism do not have same origin

• Analogous structures evolves due to convergent evolution

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• When different organism adapt to same environment

• Wings of insect and birds

• Fins of fish and penguin

• Limbs of human, horse and dolphin

• For similarity we should not only depends on general appearance

• Analogous structure act as evidence that theory of natural selection persists and adaptations accumulates over time

• An example is the convergence of the streamlined form in the bodies of squid, shark, seal, penguin, and ichthyosaur, animals of diverse ancestry.

• Physiological processes and behaviour patterns may also exhibit analogous convergence.

• Egg-guarding behaviour in the cobra, the stickleback, the octopus, and the spider is thought to have evolved independently among those animals, which are quite distant in their biological relationships.

Module# 188 Natural Enemies

• Predators

• Parasitoids

• Pathogens

Natural Enemies

• Predators kills and consume other organisms (prey)

• Usually prey is smaller in size

• Weak

WHO EATS WHO??	
PREDATORS	**PESTS ATTACKED**
Lady beetles	Aphids, mites, Lepidoptera eggs
Green lacewing	Aphids, lepidoptera eggs and small larvae
Hoverflies	Aphids
Spiders	Generalist predator, including moths, larvae, eggs, bug nymphs and other spiders
Ants	Eggs and larvae of lepidoptera
Blue birds	caterpillars, beetles, grubs
Dragonfly	Termites, mosquito larvae, other aquatic insects

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• Lion is a predators

• Spiders are predators

• Lady beetles are predators

• A parasitoid is an organism that has young that develop on or within another organism (the host), eventually killing it.

• Parasitoids have characteristics of both predators and parasites.

• Parasitoids are usually smaller than their selected host

• Pathogen is an infectious agent that feed on host body and cause damage to its host.

• Bacteria

• Viruses

• Protozoon

Module# 189 Mimicry

• Mimicry refer to similarities between species.

• Mimicry is a form of symbiosis where species may mimic colours, patterns or even behaviours for one of the two reasons:

A. Either to be able to get closer to unsuspecting prey

B. To gain protection by attaining similarity with predators or poisonous species.

• Mostly in mimic tic relationship advantage is one side.

Types of Mimicry

• Batesian mimicry

• Mullerian mimicry

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• Wasmannian mimicry

• Peckhmian mimicry

• Self-mimicry

• Batesian mimicry is a type of mimicry in which platable animal mimics the bad-tasting or noxious prey so that predator may not distinguish b/e them.

• After several attempts of distasteful encounters predators learn to avoid such appearance.

• It is called Batesian mimicry in honor of English naturalist Henry W. Bates who published an explanation of this mimicry in 1862.

Example

• Viceroy butterfly (palatable) (Mimic)

• Monarch (unpalatable/bad-taste) (Model)

Module# 190 Mimicry

Batesian Mimicry

• Eastern coral snake (poisonous)

• Scarlet King snake (non-poisonous)

• Condition for successful Batesiam Mimicry

• Number of mimics should be lower than number of models

• Mimicry breakdown

Mullerian Mimicry

• Mullerian mimicry is the form of protective mimicry in which two or more distasteful or harmful species especially insects, closely resemble each other and therefore avoided equally by all their natural predators (Muller (1821-1897)/German born Brazilian Zoologist

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• Larger the group greater will be the benefit.

Module# 191 Mimicry

Wasmannian Mimicry

• Wasmannian mimicry is a mimicry in which the mimics resembles its host in order to live within the same nest or structure.

• Several jumping spiders closely resemble ants.

• This works well in order to get "lost in the crowd" and thus avoid predators.

• Some beetles mimic ants in order to be provided with food, helter and protection.

Peckhamian Mimicry

• Peckhamian mimicry is also known as aggressive mimicry.

• In this type of mimicry predator mimics its prey in order to catch it.

• Bolas spiders are type of aggressive mimics.

• They produce chemicals that mimics the sex pheromones of particular moth species.

• When male moths approach what they perceive to be a females in order to mate with her, they are caught by bolus spiders

Module# 192 Species Concepts

Nominnalistic Species Concept

• According to this concept " Nature produces individuals and nothing more"

• It is also known as Linneaus species concept.

• Proponents of this concept (Buffon, Bessey, Lamark etc.) belive that only individuals exits but they do not believe in existence of species.

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• Species are man's own creations

• No actual existence in nature.

• Mental concept nothing more.

• Such mental concept of man has no value.

• This concept was popular in France in 18th Century.

Criticism

• Simpson (1961), Rollins (1965) and Mayr (1969) stated that no biologist can agree with the idea that species do not exist as it is established fact.

• Species are product of evolution.

Biological Species Concept

• "A species is a group of interbreeding natural population that is reproductively isolated from other such species"

• A species has three properties:

• Reproductive community

• Ecological Unit

• Genetic Unit

• Not application on asexually reproducing individuals

• Not applicable on sibling/cryptic species

• Fossil record

• Incomplete speciation

Module# 193 Species Concepts

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Species Concepts

• Cohesion species concept emphasizes the point that something infect hold the species together.

• It has been suggested that something other than sex might play this role.

• Morphological species concept

• It is based on the morphological variations.

• Variations/different species/sub-species/sibling species

Module# 194 Worm Blooded vs Cold Blooded animals

(NEXT All Lectures are VERY IMPORTANT for MCQS & SHORT QUESTIONS)

Worm Blooded animals

• They maintain their body temperature

• Examples: Birds, Mammals

Cold Blooded animals

• Do not maintain their body temperature

• Examples: Amphibians, Fishes

Problems: Some cold blooded fishes, Lizards maintain body temperature and some worm blooded animals such as birds do not

Module# 195 Ecotherms, Endotherms & Heterotherm

Ectotherms

• Produce metabolic heat at low level and quickly exchange with environment.

• Absorb heat from surroundings to maintain body temperature

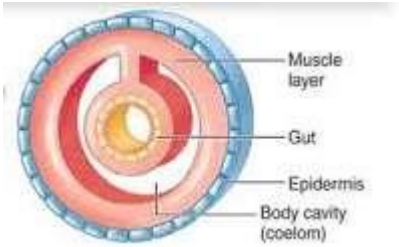
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• Examples: Mostly invertebrates, Amphibians, Fishes etc

Enthotherm



• Generate own body heat for temperature regulate

• Examples: Birds, some Ecotherms, Endotherms & Heterotherm

Heterotherms

• Capable of producing endothermic heat but generally do not use this to regulate their body temperature within a narrow range

• Examples: Bats, humming birds etc.

Module# 196 Coelomates, Acoelomates &

Pseudocoelomates: Means absences or False body Cavity.

Coelom

• Body cavity between body wall and alimentary canal

• Schizocoel/enterocoel

• Acoelomates: Coelenterates, Platyhelminthes

• Coelomates: Mollusk, Annelids, Arthropod, Echinoderms, Vertebrates etc.

• Pseudocoelomates: Nematodes

Module# 197 Protostomes Vs Deuterostomes

Protostomes

• Any member of the lower invertebrate phyla in which the mouth appears before the anus during development, cleavage is spiral and determinate and the coelom forms as a splitting of the mesoderm.

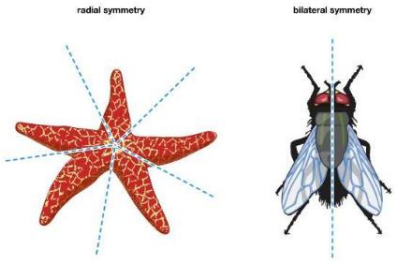
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• **Examples:** Leeches, Insects, Spiders Crustaceans, Centipedes and millipedes, Earthworms Squid, Octopus, Snail and slugs, Bivalve

Deuterostomes



• In deuterostomes, the first cavity formed by the blastopore ends up as the organism's anus, while the mouth is formed secondarily on the opposite side.

• The animals whose embryonic development undergoes radial cleavage.

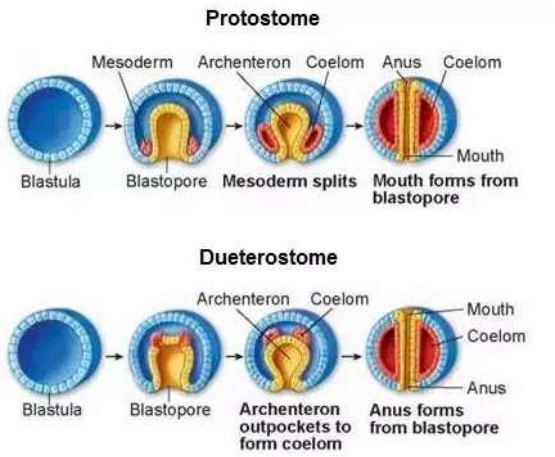
Examples of Deuterostomes

• Echinodermata

• Chordates include marine and terrestrial animals (mammals, reptiles, amphibians, frog, turtle, lizard, bat, tetrapods, Sloth, and jawless fish).

• Cephalochordata

• Vertebrata



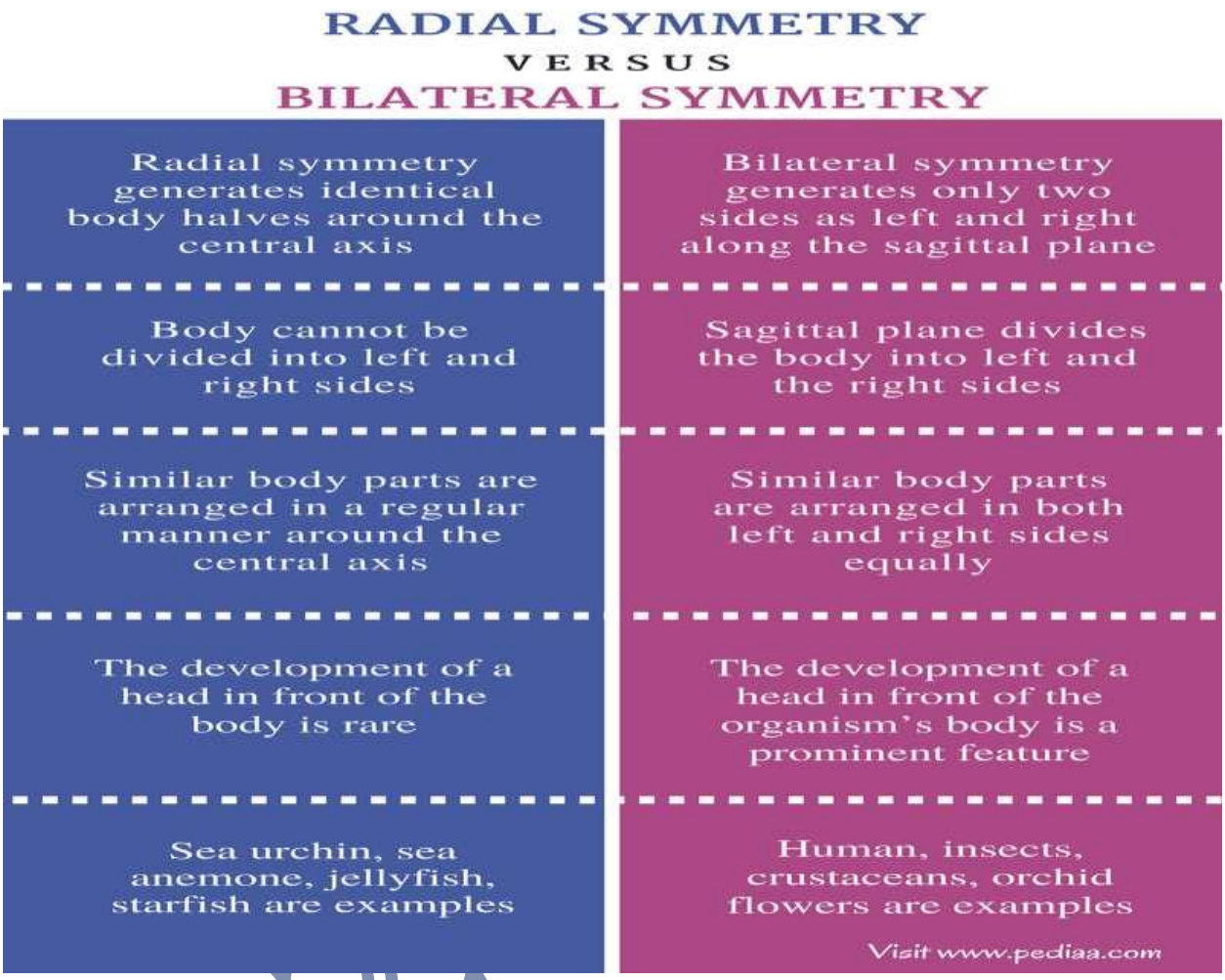
Module# 198 Radial Symmetry Vs Bilateral Symmetry

• The key difference between radial and bilateral symmetry is that radial symmetry generates identical body halves around the central axis whereas bilateral symmetry generates only two sides as left and right. • **Examples of radial symmetry** includes starfish, hydra

• **Examples of bilateral symmetry** includes insects, vertebrates

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• Activity

• Development

Module# 199 Chordates, Invertebrates Vs vertebrates

Chordates

• Five characteristic features of chordates present during some time of their life cycles are a notochord, a dorsal hollow tubular nerve cord, pharyngeal slits, endostyle/thyroid gland, and a post-anal tail.

• Invertebrates are animals with no vertebral column or backbone.

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• Invertebrates comprise approximately 97% of the animal kingdom.

• Most of them are soft- bodied and do not develop rigid internal skeletons.

• Many invertebrates do, however, possess hard exoskeletons to protect their bodies from their

• Some common examples of invertebrates are snails, sponges, earthworms, squids, sea-stars, centipedes, butterflies, spiders, and jellyfish.

• Fish

• Amphibian

• Reptiles

• Birds

• Mammals

Module# 200 Amniotes Vs Anamniotes

• Amniotes and Anamniotes are two vertebrates groups.

• Amniotes are higher vertebrates including reptiles, birds, and mammals

• Anamniotes are lower vertebrates including fishes and amphibians.

• The anamniotes are an informal group comprising the fishes and the amphibians, the so-called "lower vertebrates", which lay their eggs in water.

• They are distinguished from the amniotes, the "higher vertebrates" (reptiles, birds and mammals), which lay their eggs on land or retain the fertilized egg within the mother.

Module# 201 Diploblastic Vs Triploblastic

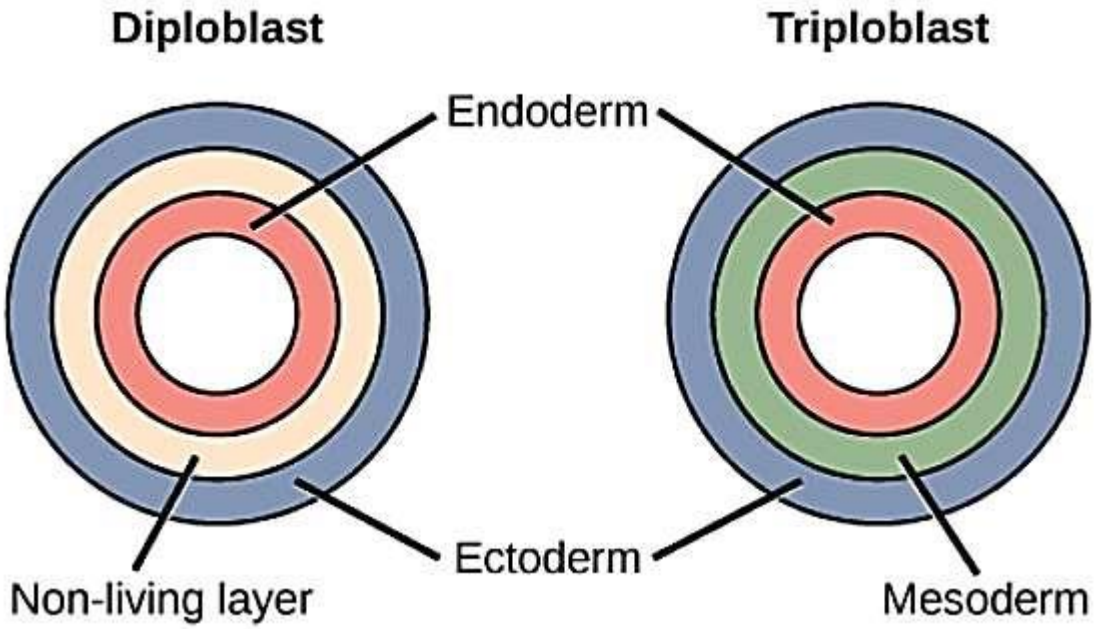
• Diploblastic organisms have only two primary germ layers in their blastula: endoderm and ectoderm.

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• They lack the middle layer or the mesoderm.



• The inner layer, which is the endoderm, gives rise to tissues associated with gut and associated glands.

• In contrast, the outer layer, ectoderm, gives rise to covering tissues like epidermis.

• Animals in phyla Cnidaria and Ctenophore belong to this group.

• Cnidarians include jellyfish, corals, sea pens, sea anemones, etc. and ctenophores include comb jellies.

• These simple, primitive metazoans basically lack body cavities and true organs.

• Most metazoans develop three primary germ layers in their blastula; hence, they are referred to as triploblastic animals.

• The three primary germ layers are ectoderm, mesoderm, and endoderm.

• Ectoderm basically gives rise to the epidermis and may also give rise to sensory organs and parts of the nervous system.

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• Mesoderm mainly forms muscles, connective tissues, blood vessels, epithelial lining of interior cavities, certain excretory organs, and skeletal elements. Endoderm gives rise to parts of gastrointestinal tracts, respiratory tract, parts of endocrine glands and organs, and auditory system.

• Examples include arthropod, Echinoderms, Mollusks, vertebrate etc.

Module# 202 Classification of Animals on the Basis of

Nutrition

• Herbivores- Animals that only eat plant. … Rabbits, Goat, sheep, grasshoppers etc

• Carnivores - Animals eat only other animals. … dog, lion

• Omnivores- Animals that eat both plants and animals.… Man, fox, Rats, crows

• Detrivores or Scavengers- Animals that eat the remains of died animal Earthworm

• Herbivores- Two important groups of herbivores are rodents (rat & squirrel) and ungulates.

• Rodents is the largest group of mammals.

• Ungulates are hoofed grassing animals such as horses, cattle and sheep.

• Premolar and Molars have large grinding surfaces.

• Canines are missing

• In grassing and browsing herbivores i.e., dear and sheep, there are no upper incisors.

Module# 203 Principles of Systematics

Classification of Animals on the Basis of Nutrition

Carnivores

• The animals which feed on other animals are called carnivores.

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• Have large canine teeth for catching and tearing the prey.

• Incisors, premolars and molars are all adapted for cutting flesh, cracking bones and reducing the chunks to sizes suitable for swallowing.

Predator

• Predator is an animal, which captures and readily kills live animals for its food.

• The animal which is eaten by predator is called prey.

• Prey-predator interaction helps marinating ecosystem stable.

• A species in the area without its natural predator leads to disastrous results.

Omnivores

• Animals which eat both plants and animals.

• They are teethes structurally and functionally intermediate between the extremes of specialization attained by the teethes of herbivores and carnivores.

• Man, Cow,

Detrivores

• Detritivores(also known as detrivores, detritus feeders, or detritus eaters) are heterotrophs that obtain nutrients by consuming detritus (decomposing plant and animal parts as well as faeces).

• There are many kinds of invertebrates, vertebrates and plants that carry out coprophagy.

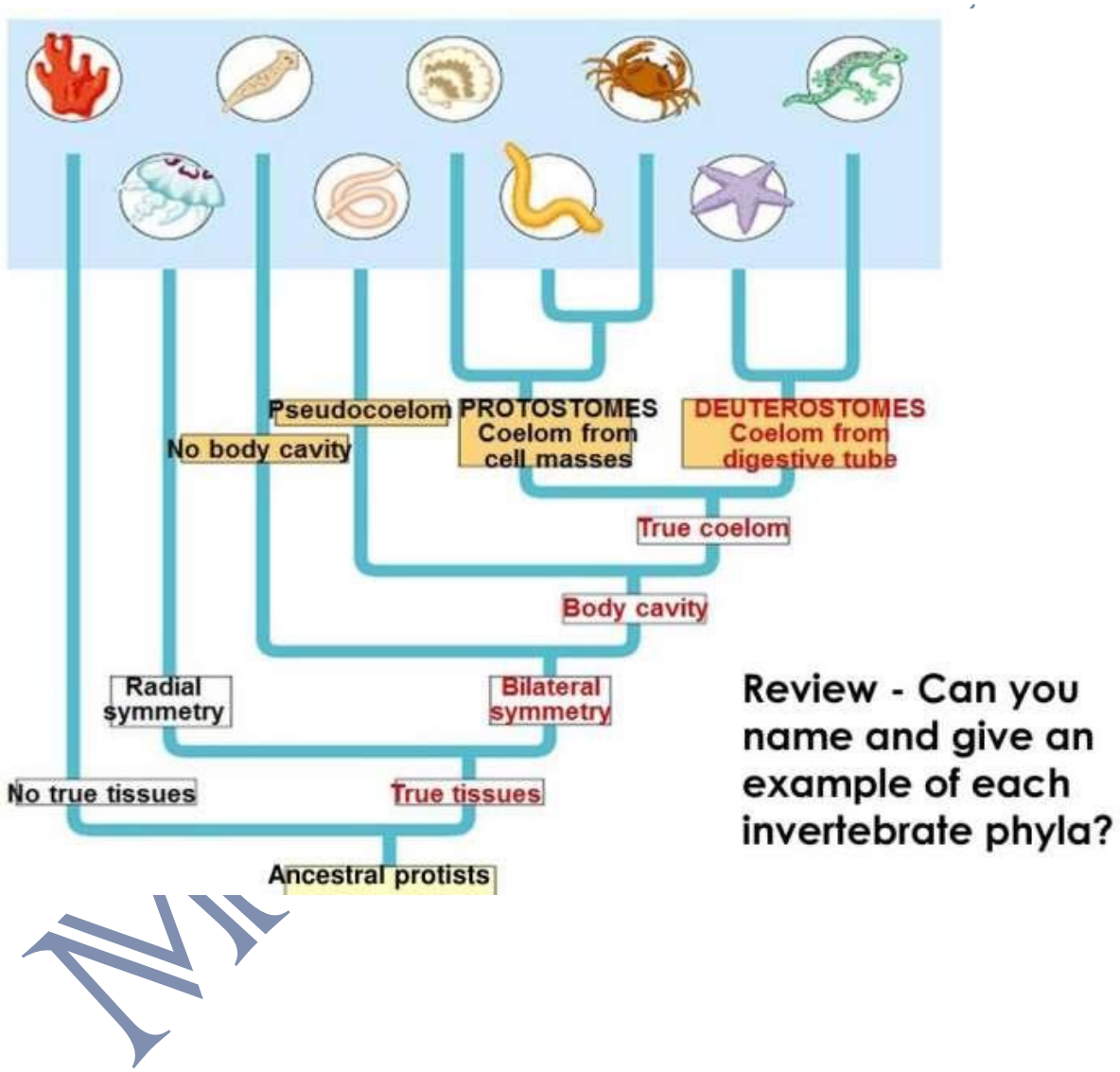
• Detritivores are often invertebrate insects such as mites, beetles, butterflies and flies; mollusks such as slugs and snails; or soil-dwelling earthworms, millipedes and woodlice.

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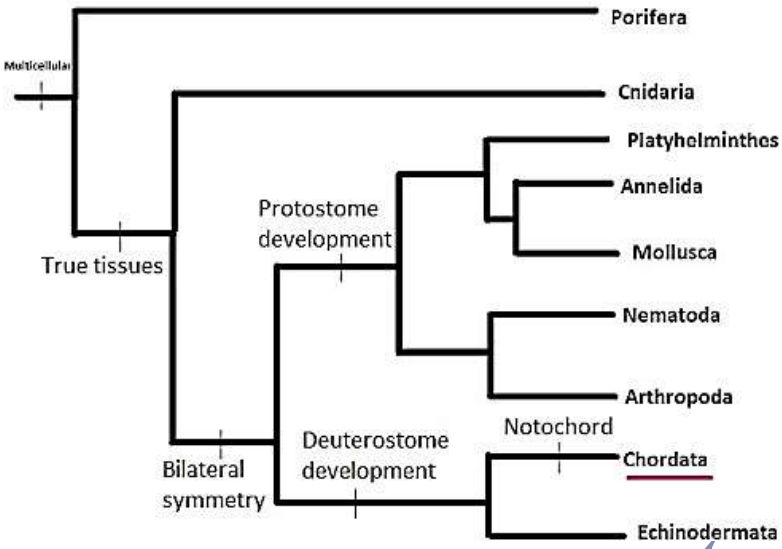
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Module# 204 Invertebrates



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Module# 205 Phylum Porifera

• Poriferans are commonly referred to as sponges. An early branching event in the history of animals separated the sponges from other metazoans.

• Fossil sponges are among the oldest known animal fossils, dating from the Late Precambrian.

• The approximately 5,000 living sponge species are classified in the phylum Porifera, which is composed of three distinct groups, the Hexactinellida (glass sponges), the Demospongia, and the Calcariea (calcareous sponges).

• Sponges are characterized by the possession of a feeding system unique among animals. Poriferans don't have mouths; instead, they have tiny pores in their outer walls through which water is drawn.

• Cells in the sponge walls filter food from the water as the water is pumped through the body and out other larger openings.

• The cells of Poriferans are loosely organized.

• They are mostly found in marine water.

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• Only a few are found in freshwater.

• They are either radially symmetrical or asymmetrical.

• Their body is usually cylindrical.

• The scleroblast secretes spicules while spongin fibres are secreted by spongioblasts.

• They have no organs in their body.

• They depict cellular grade of organization.

• The body comprises numerous pores known as Ostia and osculum.

• The central cavity is called spongocoel or atrium which opens to the outside through the osculum.

• They reproduce asexually by budding and fragmentation.

• The nutrition is holozoic.

• They have the power to regenerate the lost parts.

• The exchange of respiratory gases and nitrogenous wastes occurs by the process of diffusion.

Module# 206 Phylum Coelentrata

• Coelentrata also known as Cnidaria.

• This name of cnidaria is due to single cavity Coelenteron.

• Cnidarian have tissue but don't have organ or organ system. That's why they posses tissue level of organization.

• They are Radially symmetry and Diploblastic.



• Radial Symmetry

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• Diploblastic

• No separate body cavity.

• They have incomplete digestive system. It has a single aperture mouth but don't have anus.

• They respire through body surface by diffusion.

• They excrete through body surface by diffusion.

• Segmentations are absent in them.

• Tentacles are present in them which act as appendages.

• They are mostly marine, aquatic animals and free-living animals.

• They reproduce both sexually and asexually.

• Ex :- Sea-pan, Sea-fan, Brain coral etc.

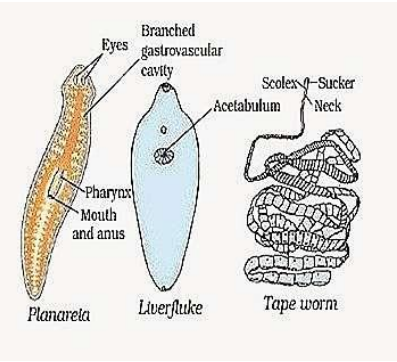
Module# 207 Phylum Platyhelminthes

• Platys: flat; helminthes: worms)

• Kingdom: Animalia

• Habitat: mostly parasitic, few are free living in sea water or fresh water

• Grade of organization: Organ-system grade of organization



• Symmetry: Bilateral symmetry and Body is dorso-ventrally flattened

• Germ layer: Triploblastic.

• Ectoderm is very specialized with distinct epidermis

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• Coelom: Absent (acoelomate).

• Digestive system: incomplete or absent

• No mouth (oral opening directly into pharynx).

• No true stomach structure (pharynx opens into a complex "intestinal" structure).

• In tapeworms, Direct absorption of soluble nutrients by cells and tissues.

• Excretory system: Protonephridia with Flame cell

• Respiration: by simple diffusion of gases through body surface

• Circulatory system: Absent

• Reproduction

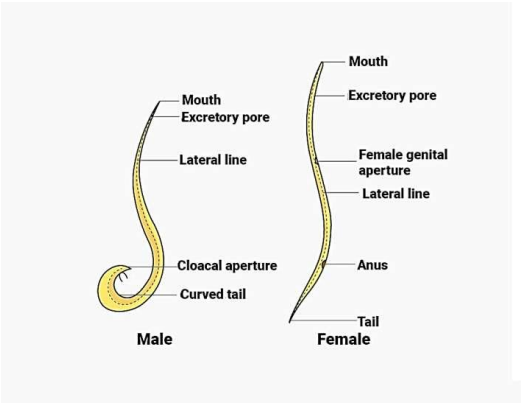
• Sexual: by gamatic fusion in hermaphrodite species

• Asexual : by regeneration and fission

• Fertilization: internal.

• Life cycle is complex involving one or more hosts.

• Examples: Tapeworms, Planaria



Module# 208 Phylum Nematodes

• Their body is bilaterally symmetrical and triploblastic.

• They are cylindrical in shape.

• They exhibit tissue level organization.

• Their body has a cavity or pseudocoelom.

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• The alimentary canal is distinct, with the mouth and the anus.

• They are sexually dimorphic.

• They are devoid of the circulatory system and respiratory system.

• They are free-living or parasitic.

• Parasitic nematodes cause diseases in the host.

• Fertilization is internal and reproduction is sexual.

• Their cuticle moults periodically.

• The epidermis is synctical and contains dorsal or ventral nerve cords.

• The body-wall muscles are longitudinal.

• They possess amoeboid sperm cells.

• They consist of chemosensory organs called aphids situated on the lips.

Module# 209 Phylum Annelida

• Their body is bilaterally symmetrical

• They have a long and segmented body.

• Annelids are bilaterally symmetrical.

• They are triploblastic.

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Nereis *Earthworm* *Leech*

Annelida

• Also, they exhibit organ system grade of organisation, showing organ differentiation

• The body is covered with a thin cuticle.

• They are coelomate.

• Annelids live in moist environments, moist soil, freshwater and marine water.

• They have parapodia and chitinous setae, used for locomotion.

• Their body appears red due to the presence of haemoglobin.

• Excretory and nervous systems are present.

• The digestive system is complete and developed.

• Respiration happens through the general body surface.

• These invertebrates have a true closed circulatory system.

• Sexes may be separate or united, wherein they are called hermaphrodites.

• Fertilization can be internal or external.

• Examples of Annelids

• Earthworm

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• Leeches

• Polychaetes

Module# 210 Phylum Mollusca

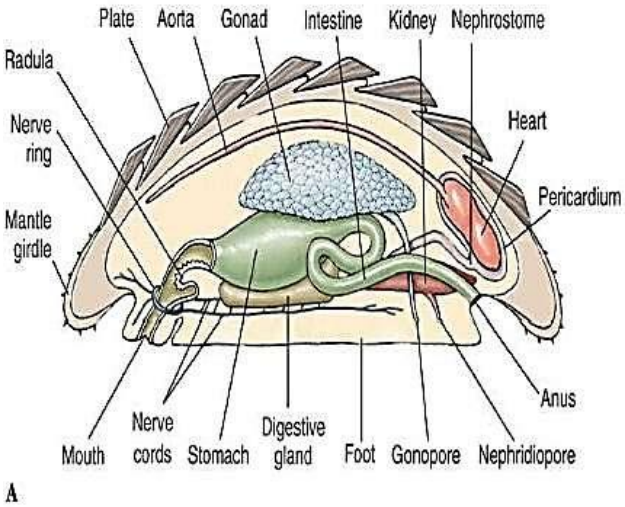
• They are mostly marine.

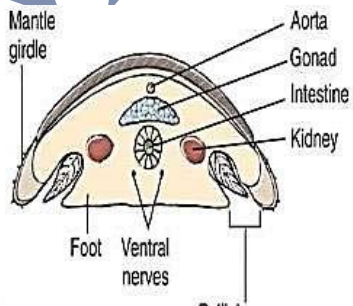
• They have segmented body with distinct head, muscular foot and visceral hump.

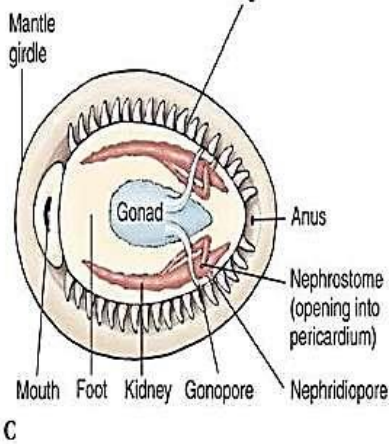
• They show bilateral symmetry.

• The shell may be external, internal or absent. It is secreted by mantle and made up of calcium carbonate.

• Coelom is greatly reduced. It is restricted to pericardial cavity and to small spaces within kidneys and gonads.







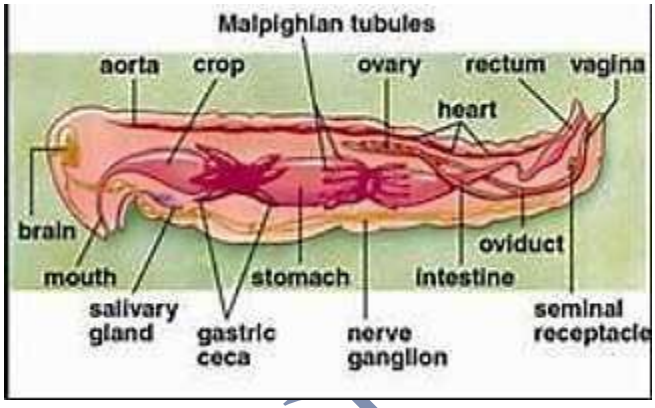
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• They have complete digestive tract.

• They have open blood vascular system.

• Blood is blue due to copper containing blue respiratory pigment called haemocyanin.



• Respirator organs are gills.

• Excretory organs are one or two pairs of sac-like kidneys.

• Osphradium is present in some molluscs for testing physical and chemical nature of water.

• The sexes are separated but some are hermaphrodite.

• Nervous system consists of cerebral, visceral, pleural and pedal ganglia.

Module# 211 Phylum Arthropoda

• Arthropods occur in all types of habitats i.e., marine, fresh water and terrestrial.

• There is an increasing tendency of cephalisation in arthropods.

• Their body is metamerically segment; composed of a linear series of segments or somites.

• Externally endoskeleton is divided into hardened segmental scletites. Internally the nervous system, muscular system and heart chambers are segmentally repeated.

• Metamerism is heteronomous.

• Appendages are joined. They act as lever systems providing mechanical advantages for locomotion.

• Muscles of arthropods are striated

• Nervous system consists of nerve ring and a ganglionated double ventral nerve cord.

• Sense organs include compound eye, simple eyes, antennae, statocyst's etc.

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• Most arthropods are unisexual. Fertilization may be external or internal in aquatic arthropods but internal in all the terrestrial forms.

• Eggs are centrolecithal (yolk in center).

• Cleavage is meroblastic and superficial.

Phylum Echinodermata

• They are exclusively marine and are among the most common and widely distributed marine animals.

• Symmetry usually radial, nearly always pentamerous.

• Body shape globular, star-like, spherical, discoidal, or elongated.

• Water -vascular system of coelomic origin, including podia or tube feet for locomotion and usually with a madreporite.

Module# 212 Phylum Echinodermata

• Sexes are usually dioecious with few exceptions.

• Gonads large and single or multiple; fertilization external, while few echinoderms are viviparous.

• Regeneration of lost parts, a peculiarity.

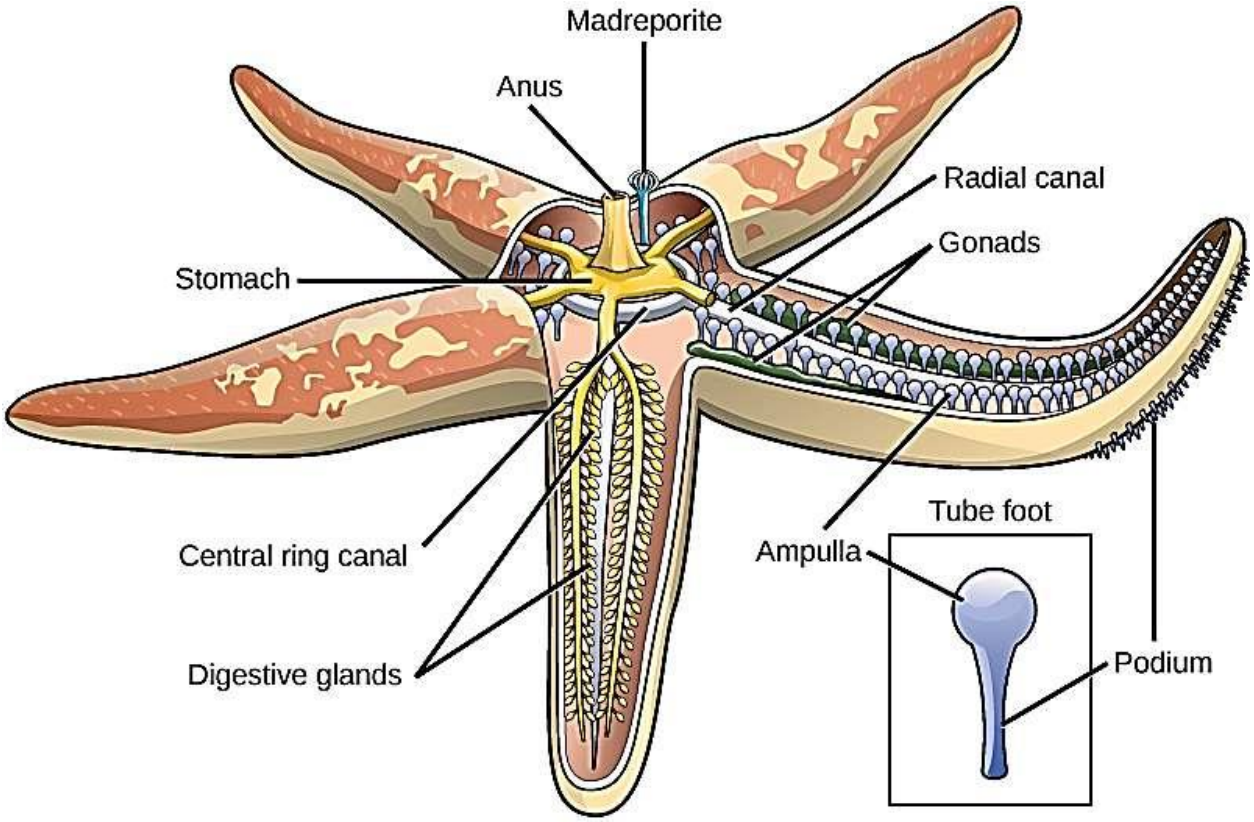
• A nervous system without a brain and with a circumoral ring and radial nerve.

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• The excretory system is wanting.



• Respiratory organs include branchiae, tube-feet, respiratory tree, and bursae.

• Development is intermediate including characteristic larvae which undergo metamorphosis into the radially symmetrical adults.

Phylum Echinodermata

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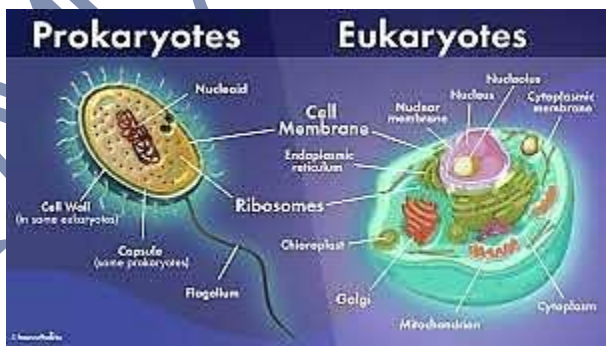
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Module# 213 Prokaryotes Vs Eukaryotes

• Prokaryotes are organisms made up of cells that lack a cell nucleus or any membrane-encased organelles.

• Eukaryotes are organisms made up of cells that possess a membrane-bound nucleus that holds genetic material as well as membrane-bound organelles.



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	Prokaryotes	Eukaryotes
DNA	DNA is naked	DNA bound to protein
	DNA is circular	DNA is linear
	Usually no introns	Usually has introns
Organelles	No nucleus	Has a nucleus
	No membrane-bound	Membrane-bound
	70S ribosomes	80S ribosomes
Reproduction	Binary fission	Mitosis and meiosis
	Single chromosome (haploid)	Chromosomes paired (diploid or more)
Average Size	Smaller (~1–5 µm)	Larger (~10–100 µm)



Prokaryote vs eukaryote: key differences

	Prokaryote	Eukaryote
Nucleus	Absent	Present
Membrane-bound organelles	Absent	Present
Cell structure	Unicellular	Mostly multicellular; some unicellular
Cell size	Smaller (0.1-5 µm)	Larger (10-100 µm)
Complexity	Simpler	More complex
DNA Form	Circular	Linear
Examples	Bacteria, archaea	Animals, plants, fungi, protists

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Module# 214 Class Pisces

• They are found in fresh, marine, and brackish water.

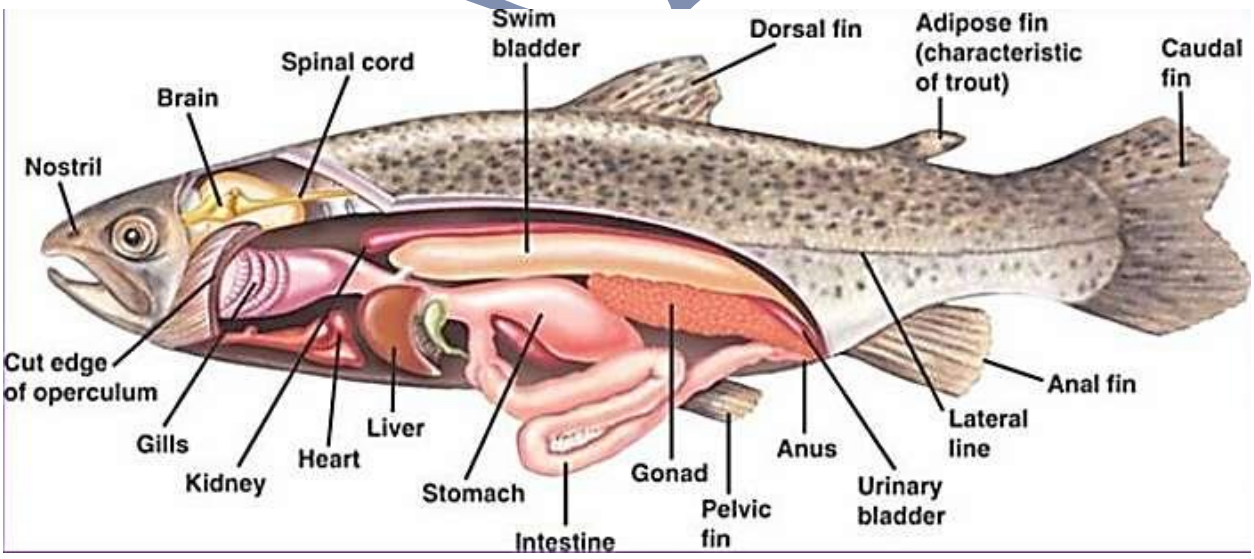
• The body is usually streamlined. Some have a spindle-shaped or elongated body as well.

• Their body is distributed into a head, trunk and tail.

• They swim with the help of their tail.

• Paired and unpaired fins represent the appendages. These help the fish to balance while swimming.

• The lateral line system functions as a sensory organ to sense the disturbances in the nearby environment.



• The body is covered with thick- seated scales, which helps by providing protection to the internal organelles.

• The gills help in respiration.

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• Closed type blood circulation is observed.

• The internal skeleton is bony or cartilaginous.

• These are cold-blooded organisms.

• Swim bladder

• They may be herbivores or carnivores, oviparous or ovoviviparous.

• The sexes are separate.

• Fertilization may be external or internal.

• They lack extra-embryonic membranes.

• The digestive system is well- developed.

• The nervous system comprises of the brain and ten pairs of the cranial nerves.

• Types

Module# 215 Class Amphibians

• They are the first cold blooded vertebrates from evolution point of view which came to the land.

• They occur in fresh water and moist land.

• Body is divisible into head and trunk. Tail may be present in some amphibians.

• The skin is smooth or rough having glands which keep it moist.

• They are usually without scales, but if present they are hidden beneath the skin (e. g. caecilians).

• Paired fins are absent. Unpaired fins may be present. Two pairs of limbs are used for locomotion except caecilions.

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• The gills are present at least in the larval stage; some adult forms also carry external gills in addition to lungs (e.g., Proteus).

• Skull is dicondylic, i.e., with two occipital condyles for articulation with vertebral column.

• The respiratory organs are lungs, buccopharyngeal cavity, skin and gills.

• The heart is three chambered, having two auricles and one ventricle.

• Kidneys are mesonephric.

• Alimentary canal, urinary and reproductive tracts open into a common chamber called cloaca which opens outside through cloacal aperture.

• Ear consists of internal and middle ear. Tympanum (outer membrane) covers the middle ear. The eyes have eyelids. Nictitating membrane is well developed.

• Ten pairs of cranial nerves are present.

• Lateral line system is found during their development.

• Fertilization is external.

• Reptiles are the creeping and burrowing cold blooded vertebrates bearing epidermal scales.

• They are ectothermic (cold- blooded) and are found mostly in the warmer parts of the world.

• Skin is dry, rough and without glands, bearing epidermal scales or scutes.

• Snakes and lizards shed their scales as skin cast.

Module# 216 Class Reptilia

• They do not respire by means of gills. Respiration always takes place through lungs.

• Skull is monocondylic, i.e., with single occipital condyle.

• Except in snakes, there are two pairs of pentadactyl limbs, each with 5 digits bearing claws—tetrapodus pentadactyl type.

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• Heart consists of two auricles and a partially divided ventricle. In crocodilians, heart is four chambered (two auricles and two ventricles).

• Twelve pairs of cranial nerves are present.

• Each ear consists of three parts; external, middle and internal. Snakes do not possess ears.

• The lateral line system is altogether absent.

• Tortoises feed almost entirely on vegetation.

• Some turtles are flesh eaters.

• All other reptiles are carnivorous/insectivorous.

• A typical cloaca is present.

• They are mostly oviparous.

Module# 217 Class Aves

• Birds are bipedal feathered and warm blooded (homoiothermous) animals.

• The hind-limbs are adapted for perching, walking or swimming, etc., and usually bear four, sometimes three and rarely two toes.

• Except uropygium gland (preen gland or oil gland), at the base of the tail, no skin gland is present. Ostrich and parrot lack oil gland.

• The upper and lower jaws are modified into beak, which lacks teeth.

• Legs are modified for walking, hopping, grasping, perching, wading and swimming. Legs bear homey epidermal scales.

• The alimentary canal has additional chambers, the crop and gizzard.

• The crop stores and softens the food; however, the gizzard helps in crushing and churning the food.

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• Gall bladder is absent in some seed-eating birds (graminivorous) such as pigeons.

• Respiration is by lungs. The lungs are spongy and inelastic. Air sacs are connected to lungs for supplement respiration.

• Voice is produced by a special organ, the syrinx.

• The heart is four-chambered. Sinus venosus is absent.

• Renal portal system is very much reduced. Red blood corpuscles are nucleated, oval and biconvex.

• The kidneys are metanephric.

• Birds have 12 pairs of cranial nerves.

• Female has usually well-developed single left ovary and oviduct. If right ovary and oviduct are present, they are vestigial (nonfunctional).

• Many birds show sexual dimorphism.

• All birds are oviparous.

• They show courtship, nest building, parental care, migrationand territorial behaviour.

Module# 218 Class Mammalia

Class Mammalia

• These animals are warm blooded, hairy and have mammary or milk producing glands, (mammary glands).

• They are homoiothermous.

• Oil glands (sebaceous glands) and sweat glands (sudoriferous glands) are present in the skin.

• Teeth are of different types (heterodont) and are embedded in the sockets of jaws (the codont).

• Except a few, mammals possess seven cervical (neck) vertebrae.

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• The skull is dicondylic i.e., with two occipital condyles.

• Respiration is by lungs.

• They possess a muscular diaphragm dividing trunk into thorax and abdomen.

• The coelom is divided into four cavities.

• A pericardial cavity lodging the heart, two pleural cavities each containing the lung and an abdominal cavity having the rest of viscera.

• The heart is four chambered. Sinus venosus is absent.

• The red blood corpuscles are without nucleus.

• Renal portal system is absent.

• The brain has large cerebrum and cerebellum. Optic lobes are divided into four lobes called corpora quadrigemina.

• Corpus callosum connects the two cerebral hemispheres internally.

• 12 pairs of cranial nerves are present.

• Each ear consists of three parts: external, middle and internal. Pinna is a part of external ear. Middle ear has 3 bony ear ossicles.

• Except egg laying mammals they are viviparous.

• They are dominant animals and are capable to learn because of their better developed brain.

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Jazak Allah

Effort By M Zaman

“Don’t Stop when you Tired, STOP when you DONE”

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